

THE
JOURNAL OF GEOLOGY
May-June, 1893

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THE JOURNAL OF GEOLOGY

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ON THE TYPICAL LAURENTIAN AREA OF CANADA.

The name Laurentian was given by Logan in 1854 to the great series of rocks forming the Laurentides or Laurentian Mountains, a district of mountainous country rising to the north of the River and Gulf of St. Lawrence, and extending in an unbroken stretch along the shore of the latter from Quebec to Labrador, a distance of nine hundred miles. This district, with its continuation to the west as far as Lake Huron, being situated in the Province of Quebec and the adjacent portion of the Province of Ontario, and forming part of the main Protaxis of the continent, is the “Original Laurentian Area” of Logan. The Laurentian rocks are now known to extend far beyond the limits of this area to the west and north, constituting, as they do, by far the greater part of the Protaxis, and underlying (with subordinate patches of Huronian) an area of somewhat over two million square miles.¹ The area above referred to is, however, the one which was first studied and described; it is the “Typical Laurentian area,” and to it the observations in the present paper will be as far as possible confined.

A general exploration of the area in question, and a more detailed study of a small part of it—the Grenville District—situated in the counties of Argenteuil and Terrebonne in the Province of Quebec, was carried out by Logan and his assistants in the early years of the Canadian Geological Survey. An excellent résumé of the results of these studies is given in the “Geology of Canada,” published in 1863, which contains not only a good description of the general petrographical character and arrangement of the rocks which make up the area, but is accompanied by an atlas containing two maps illustrating this description, one showing the general distribution of the Laurentian in the eastern part of the Dominion, and the other its stratigraphical relations in the smaller area above referred to.

As a result of these studies, Logan announced his belief that the Laurentian System consisted of two great unconformable series of sedimentary rocks, to which he gave the names Upper and Lower Laurentian. The latter he considered to be divisible into a lower and an upper portion, which sub-divisions he regarded as probably conformable to one another. In the course of time these several series came to be known as the Anorthosite or Norian Series, the Grenville Series and the Fundamental or Ottawa Gneiss. Logan's views may then be represented as follows:

Anorthosite or Norian Series,	Upper	
	Laurentian.	
Grenville Series,	Upper portion	
Fundamental or Ottawa	Lower portion	} Lower
Gneiss,		
		Laurentian.

Subsequently, in the southeastern corner of the Province of Ontario, in the district lying to the north of the eastern end of Lake Ontario, another series of rocks was discovered—the so-called Hastings Series. Logan supposed this to come in above the Grenville Series, while Vennor, who subsequently examined the district, believed it to be equivalent to the lower part of the Grenville Series already mentioned.

When these investigations were carried out, the microscope had not as yet been seriously employed in petrographical work. The precise composition of many of the rocks making up the several series was not recognized, the effects produced by great dynamic action were not duly considered, and the foliation possessed in a high degree by some and to a certain extent by almost all these rocks was considered, in all cases, to be a more or less obliterated survival of original bedding. The detailed mapping in the field, accompanied by microscopical work in the laboratory, by which alone conclusive results can be obtained in working out the structure of complicated areas of crystalline schists, was not carried out, in fact in many districts the construction of detailed maps was at that time practically impossible. It is not surprising therefore that, although excellent in the main, some of the results arrived at have since proved to be erroneous.

It is proposed, in the present paper, to place before the readers of this JOURNAL in as brief a manner as possible, a general account of the several series of rocks occurring in this area, and to point out what, in the opinion of the present writer, seems to have been satisfactorily established concerning the stratigraphical position and mutual relations of these ancient rocks and what still remains to be determined by further study, and in conclusion to give a short sketch of the evolution of this portion of the continent.

The Fundamental Gneiss.—Exposed over very wide stretches of country in Canada, and making up in all probability by far the larger part of the Archean Protaxis, is the “Fundamental Gneiss,” sometimes called, from its great development about the upper waters of the Ottawa River, the “Ottawa Gneiss.” It is composed essentially of orthoclase gneiss, usually reddish or greyish in color. Of this there are a number of varieties, differing from one another in size of grain, relative proportion of constituent minerals and in the distinctness of the foliation or banding. It is sometimes rich in quartz, while at other times this mineral is present in but very small amount. It is usually poor in mica and bisilicates. Dark bands of amphibolite are not uncommon, while basic hornblende or pyroxene gneisses occur in some places. Other schistose rocks are rarely found. Over great areas it is often nearly uniform in character and possesses a foliation which can only be recognized when exposures of considerable size are examined. On this account it is often referred to as a granitoid gneiss, a designation, however, which by no means accurately describes it as a whole. At a locality cited by Sir William Logan, as one where it is typically developed, namely, Trembling Mountain in the above mentioned Grenville Area, it consists of a fine grained reddish orthoclase gneiss, with distinct but not very decided foliation, containing here and there bands of orthoclase gneiss of somewhat different character, as well as bands or layers of a dark amphibolite.

How much of this Fundamental Gneiss really consists of eruptive material is not known. The indistinct foliation, in many cases at any rate, is not a survival of original bedding, but is clearly due to movements in a plastic mass. It is often possible to recognize the existence of an indistinctly foliated gneiss intruded into more distinctly foliated gneiss. The gneiss, in some cases, shows excellently well-marked cataclastic structure, while in other cases this is not distinct. The evidence accumulated goes to show that

the Fundamental Gneiss consists of a complicated series of rocks of unknown origin, but comprising a considerable amount of material of intrusive character.

The Grenville Series.—In certain parts of the Laurentian area, and notably in the Grenville district before mentioned, the Laurentian has a decidedly different petrographical development. Orthoclase gneiss is still the predominating rock, but it presents a much greater variety in mineralogical composition, and is much more frequently well foliated, often occurring in well defined bands or layers like the strata of later formations.

Amphibolites are abundant, also hornblende schists, heavy beds of quartzite and numerous thick bands of crystalline limestone or marble, all these rocks being interbanded or interstratified with one another. In the vicinity of the limestones the variety in petrographical character is especially noticeable; garnets often occur abundantly in the gneiss, the quartzite and the hornblende schist, as well as in the limestone itself, beds of pure garnet rock being found in places. Pyroxene, wollastonite and other minerals are also abundant, while the presence of graphite disseminated through the limestones and their associated rocks, often in such abundance as to give rise to deposits of economic value, is of especial significance. This mineral which is not found in the Fundamental Gneiss, occurs usually in little disseminated scales but occasionally in veins. The limestones are thoroughly crystalline, generally somewhat coarse in grain and often nearly pure. They usually, however, contain grains of serpentine, pyroxene, mica, graphite or other minerals, of which over fifty species have been noted. They are often interstratified in thin bands with the gneiss, in places are very impure, and may be traced for great distances along the strike, being apparently as continuous as any other element of the series. This development of the Laurentian is known as the Grenville Series, and has been considered by all observers to be above and to rest upon the Fundamental Gneiss. In it are found all the mineral deposits of economic value—apatite, iron ore, asbestos, etc., which occur in the Laurentian. The rocks of this series, though generally highly inclined, over some large areas lie nearly horizontal or are inclined at very low angles, but even in such cases they show evidence of having been subjected to great pressure, resulting in some cases in the horizontal disruption of certain of the beds.

The areas occupied by the Grenville series although of very considerable extent, being known to aggregate many thousand square miles, are probably small as compared with those underlain by the Fundamental Gneiss. The relative distribution of the two series has not been ascertained except in a general way in the more easily-accessible parts of the great Archean Protaxis. The Grenville series is known to occupy a large part of its southern margin between the city of Quebec and the Georgian Bay, while the discovery of crystalline limestone in the gneiss elsewhere at several widely separated points, as for instance on the Hamilton River in Labrador, in the southern part of Baffin Land and on the Melville Peninsula, makes it probable that other considerable areas will, with the progress of geological exploration, be found in the far north. Over the greater part of the Protaxis, however, the more monotonous development of the Fundamental Gneiss seems to prevail.

The question of the origin and mutual relations of the Fundamental Gneiss and the Grenville series is one about which, though much has been written but little is known. Three views may be taken on the matter—

(1) The Fundamental Gneiss may be supposed to contain what remains of a primitive crust, penetrated by great masses of igneous rock erupted through it—the whole having been subjected to repeated dynamic action.² The Grenville Series may be an upward continuation or development of the Fundamental Gneiss under altered conditions, marking in the history of the world the transition from those conditions under which a primitive crust formed to those in which sedimentation under the present normal conditions took place. It would seem that if the earth originally had a crust on which the first sediments were deposited when the temperature became sufficiently low to permit water to condense, that the said water, at a very high temperature and under what are to us now inconceivable conditions but little removed from fusion, might give rise to sediments not altogether similar to those formed by the ordinary processes of erosion at the present time. Also that, under the unique conditions which must have prevailed at that early time, in the formation of a crust solidification, precipitation and sedimentation might go on to a certain extent concomitantly, and thus no well-defined break could be detected, or would in fact exist, between a primitive crust formed by solidification from a fused magma and the earliest aqueous sediments or deposits. The Fundamental Gneiss and the

Grenville Series might thus, as Logan supposed, form one practically continuous series and represent parts of the original crust, with the first crystalline or clastic sediments deposited on it, the whole penetrated by eruptive rocks and folded up and altered by repeated dynamic action at subsequent periods.

The general petrographical similarity of the two series, taken in connection with the more varied nature of the Grenville Series, its frequent stratified character, and the presence in it of limestones and graphite indicating an approach to modern conditions and the advent of life, together with the difficulty of clearly separating the two series from one another and defining their respective limits, lends support to this view.

(2) A second view is that the Grenville Series is distinct from the Fundamental Gneiss reposing on it unconformably and of much more recent age; that it consists of a highly altered series of clastic origin—the Fundamental Gneiss having possibly some such origin as that mentioned under the last heading, or representing a much older series of still more highly altered sediments. This is supported by the fact that some observers have thought they could in places trace out a line of contact between the two. But in these cases it always becomes a matter of serious doubt whether what has been considered to represent the Fundamental Gneiss is not really a mass of intrusive rock, in which, by pressure or motion, a somewhat gneissic structure has been induced. If the Fundamental Gneiss, moreover, was ever an ordinary sediment, it must have undergone a metamorphosis so profound that no trace of clastic origin remains, unless the generally indistinct foliation or banding of some portions of it be considered as such. It must also be noted in this connection that, although the rocks of the Grenville series are more frequently possessed of a decided foliation and are often banded, bands of different composition alternating with one another as in ordinary sedimentary deposits, and although in this series crystalline limestones and quartzites occur, we have as yet no absolutely conclusive proof that even they are of sedimentary origin. The series is thoroughly crystalline, most of its members at least show the effect of great dynamic action, and so far as the present writer is aware, no undoubted conglomerate or finer grained rock showing distinct clastic structure has ever been found. In view of this fact,—although the series is, in all probability, made up in part at least and perhaps wholly of sedimentary

material,—the proposal to separate it from the rest of the Laurentian and class it as Algonkian or Huronian seems at least premature.

(3) A third view which has been advanced is that the Fundamental Gneiss is nothing more than a great mass of eruptive granite or granitic rock which has eaten upward, and in places penetrated the Grenville series, or perhaps absorbed it, while the Grenville series itself represents a series of highly altered sediments of Laurentian, Huronian or subsequent age. The enormous extent and world-wide distribution of the Fundamental Gneiss forming as it does wherever the base of the geological column is exposed to view, the foundation or floor on which all subsequent rocks are seen to rest, is opposed to this view of its origin, as is also its persistent gneissic or banded character, although, as above mentioned, much eruptive material is undoubtedly to be found in it.

Which of these views is correct can be ascertained only as very careful and detailed mapping, accompanied by accurate petrographical study, is proceeded with. In the present state of our knowledge additional argument and discussion will not help us toward the goal, while hasty work and generalization serves but to retard the progress of our knowledge.

The Anorthosite Series.—Associated with both the series of rocks just described there are, as has been mentioned, great eruptive masses of granite, some of which have been folded in with the gneisses, while others evidently erupted at a much later date, show no trace of dynamic action.

In addition to these, basic eruptive rocks belonging to the gabbro family occur in certain districts, sometimes in the form of comparatively insignificant masses, but elsewhere underlying great tracts of country. One on the upper waters of the Saguenay has an area of no less than 5,800 square miles. These usually consist of a variety of gabbro in which the magnesia-iron constituents are present in very small amount, being in many cases entirely wanting, so that the rock consists practically of pure plagioclase feldspar. These rocks were called *anorthosites* by Hunt, in the early reports of the Canadian Geological Survey, on account of the great preponderance in them of “Anorthose,” a general name given many years ago by Delesse to the triclinic feldspars, as distinguished from “Orthose,” or orthoclase feldspar, and thus equivalent to the term plagioclase now in general use, but having no connection with anorthite, a variety of

plagioclase which is seldom present. After a careful study of these rocks, both in the field and the laboratory, it is believed that this name should be retained for this well-marked member of the gabbro family, which, though not a common rock elsewhere, has an enormous distribution in the Laurentian of Canada.

If an olivine gabbro be regarded as the central member, so to speak, of the gabbro family, the replacement of the monoclinic by rhombic pyroxene will give rise to an olivine norite. A gradual diminution in the amount of plagioclase will give rise to a peridotite or gabbro pyroxenite, a diminution in the amount of pyroxene to a troctolite or plagioclase-olivine rock, while a diminution in the amount of olivine and pyroxene will give rise to an anorthosite, which variety forms the greater part of the intrusive masses in question. The gradual passage of one variety into another can be distinctly traced in many localities in the anorthosite masses. These anorthosites are in some places massive, but very frequently show a distinct foliation, often very perfect. In some places they occur interbanded with the gneiss and crystalline limestone, while elsewhere they cut directly across the strike of these rocks. The interbanded anorthosite, together with the gneiss and limestone associated with it, was supposed by Logan to form a distinct sedimentary series, to which the name "Upper Laurentian," or "Norian," was given, because the discovery that elsewhere the anorthosite runs across the strike of the gneiss was supposed to indicate that this series covered up and unconformably overlay the Grenville series, the igneous and intrusive character of the anorthosite not being recognized on account of its frequently foliated structure. It is now known that these anorthosites do not constitute an independent formation, but are igneous rocks which occur, cutting both the Grenville series and the Fundamental gneiss. They have, however, in many cases been intruded before the cessation of the great dynamic movements to which the Laurentian was subjected in pre-Cambrian times, and thus frequently taking a line of least resistance and having been intruded between the bands or strata of the Grenville series, have had a foliation induced in them parallel to that of the gneiss, while in other cases where they are more or less massive, they cut across the strike of the latter.

In many cases the anorthosites which exhibit a perfect foliation may be traced step by step into the massive variety, the gradual development of a

foliated structure in the rock being accompanied by a progressive granulation of the constituents, most beautifully seen under the microscope. The change, however, differs from any hitherto described in that it is purely mechanical. There are no lines of shearing with accompanying chemical changes, but a breaking up of the constituents throughout the whole mass, though in some places this has progressed much further than in others, unaccompanied by any alteration of augite or hypersthene to hornblende, or of plagioclase to saussurite, these minerals, though prone to such alteration under pressure remaining quite unaltered, suffering merely a granulation with the arrangement of the granulated material in parallel strings. This process can be observed in all its stages, and there is reason to believe that it has been brought about by pressure acting on the rocks when they were deeply buried and very hot.³ The anorthosite areas, of which there are about a dozen of great extent with many of smaller size, are distributed along the south and southeastern edge of the main Archean Protaxis from Labrador to Lake Champlain, occupying in this way a position similar to that of volcanoes along the edge of our present continents. Curiously enough precisely similar occurrences of this anorthosite have been found in connection with similar gneissic rocks, supposed to be of Archean age, in Russia, Norway and Egypt. These anorthosite rocks being intrusive, may be left out of consideration in endeavoring to work out the succession of the Archean in this great area.

The whole Laurentian system, including the anorthosites, is in many places cut by numerous dykes of large size, which can often be traced for great distances. These are of several kinds, the principal series consisting of a beautiful fresh diabase often holding quartz in considerable amount in micro-pegmatitic intergrowths with plagioclase. Other sets of dykes and eruptive masses consisting of augite and mica syenites, quartz-porphyrines and other rocks are also known to occur but have not as yet been carefully studied.

The Hastings Series.—The stratigraphical relations of the Hastings series have not as yet been satisfactorily determined. The rocks constituting the series differ widely in petrographical character from those of the Fundamental Gneiss and the Grenville series, both of which are supposed to occur in its immediate neighborhood. The series consists largely of calc-schists, mica-schists, dolomites, slates and conglomerates, thus containing

much material of undoubtedly clastic origin. It has moreover a very local development, being confined, so far as at present known, to one small corner of the area, as has been mentioned. It was by Logan supposed to come in above the Grenville series, while Vennor who subsequently examined the district, believed it to be equivalent to the lower part of this series. That we have in the Hastings series a comparatively unaltered part of the Grenville series, made up largely of rocks whose origin is easily recognized, would be a most important fact if established, and would, of course, afford a key to the whole question of the origin of the latter. This is a conclusion, however, which cannot be accepted until supported by very clear and decisive evidence, especially as the stratigraphy of the Hastings district is very complicated, the several series represented in it being much folded and penetrated by great masses of eruptive rocks. The whole district has also been subject to great dynamic action, some of the pebbles in the conglomerates of the Hastings series being distorted in a most remarkable manner. This series may prove to be merely an outlying area of Huronian rocks folded in with the Laurentian, and until the district has been studied in detail its stratigraphical position must remain a matter of conjecture.

Leaving the Hastings series out of consideration therefore, we have in this Original and Typical Laurentian area two developments of the Laurentian, generally considered as constituting two series, namely the

Grenville or Upper series,
Fundamental, Ottawa, or Lower Gneiss.

The Evolution of the Area.—In endeavoring to outline the main events in the evolution of this area it will be necessary to extend the limits of our observation somewhat and seek for evidence bearing on the question in other parts of the Protaxis, where we meet with developments of Huronian and various earlier Paleozoic strata not found in the typical area itself.

From the highly contorted condition of the Laurentian rocks of this area as well as from the abundant evidences of dynamic action which they present both in the field and under the microscope, it is evident that they have been subjected to great orographic forces, which in very early times threw them up into mountain ranges, probably of great height. Some of the associated eruptive rocks were intruded before these movements began, or

while they were in progress and have accordingly been influenced by them, while others, having been intruded later, have not been affected.

How high these mountains rose cannot of course be determined. Bell states that some of the mountains on the Labrador coast now rise to a height of from 5,000 to 6,000 feet, while Lieber has estimated that on the coast of Northern Labrador they rise to a height of from 6,000 to 10,000 feet. Along the southern part of the Protaxis, where the country is much lower, notwithstanding the enormous subaerial denudation and glaciation which the area has repeatedly undergone, there are many points still rising from 2,500 to 3,500 feet above sea level, while Logan estimated that the average elevation is from 1,500 to 1,600 feet. In the Adirondacks, which are but an outlying portion of this area, there are elevations of over 5,400 feet. The high elevations attained by these rocks in portions of the Protaxis in the north may, of course, be due to differential elevation, but immediately along the southern edge of the area there can have been but little differential change of level as compared with the flat-lying Potsdam strata which border it and lie but little above the present sea level. Further evidence of the original height or continued uprising of the area is afforded by the fact that all the material of which the North American continent was built up (with the possible exception of some of the limestones) was derived originally from the Archean Protaxis of the continent, a considerable proportion of this at least coming from the main Protaxis of which this typical Laurentian area forms a part. We must conclude therefore that in early Cambrian or pre-Cambrian times, in portions of the Protaxis at least, the Laurentian mountains rose several hundred and possibly in places several thousand feet above the sea level.

The intrusion of the granites and anorthosites as well as the folding of the whole system of rocks took place before Upper Cambrian times. The whole series was moreover without doubt at that time in the "metamorphic" condition in which we now find it, for along the margin of the area the Potsdam sandstone rests in flat undisturbed beds on the deeply eroded remnants of these old mountains, its basal beds often consisting of a conglomerate with pebbles of the underlying gneissic rocks. These Cambrian strata cover up the gneisses, granites and anorthosites alike and are evidently of much more recent age, being separated from the Laurentian

by the long interval occupied in the upheaval and erosion of the Laurentian area.

How long before Upper Cambrian times this folding and erosion took place cannot be determined from a study of this area, but further west along the edge of the Protaxis in the Lake Superior district we find that the Keweenawan, Nipigon and Animikie Series also repose in flat undisturbed beds on the eroded remnants of a series of crystalline rocks which have the petrographical character of the Fundamental Gneiss. This makes it at least very probable that in this eastern area also the erosion took place in pre-Cambrian times.

It is a very remarkable fact that the *roche moutonnée* character possessed by these eroded Laurentian rocks and which is usually attributed to the glaciation which they underwent in Pleistocene times, was really impressed upon them in the first instance in these pre-Cambrian times, for all along the edge of the nucleus from Lake Superior to the Saguenay, the Paleozoic strata, often in little patches, can be seen to overlie and cover up a mammillated and *roche moutonnée* surface showing no traces of decay and similar to that exposed over the uncovered part of the area. The conclusion therefore seems inevitable that not only were these Laurentian rocks sharply folded and subjected to enormous erosion, but that they had given to them in pre-Cambrian times their peculiar hummocky contours so suggestive of ice action.⁴ The pre-Paleozoic surface of the Fundamental Gneiss of Scotland, as Sir Archibald Geikie has shown, also presents the same hummocky character.⁵ On this surface the Upper Huronian, Cambrian, and later Paleozoic rocks were deposited.

To what extent the seas of Cambrian, Silurian and Devonian times passed over this area cannot be determined with certainty. A great series of rocks referred to by Dr. G. M. Dawson as probably of Lower Cambrian age and analogous in character to the Keweenawan and Animikie series occur overlying the Laurentian in many parts of the Protaxis, not only along its margin, but as outliers at many places in the interior. It occurs extensively developed about the Arctic Ocean and about Hudson's Bay, and a large area of rocks referred to the same age also occur near the height of land about Lake Mistassini. "Throughout the whole of the vast northern part of the continent this characteristic Cambrian formation, composed largely of

volcanic rocks, apparently occupies the same unconformable position with regard to the underlying Laurentian and Huronian systems. Its present remnants serve to indicate the position of some of the earliest geological basins, which from the attitude of the rocks appear to have undergone comparatively little disturbance. Its extent entitles it to be recognized as one of the most important geological features of North America.”⁶ It would, therefore, seem that in Cambrian times a not inconsiderable part of the Archean Nucleus was under water. Outliers of Cambro-Silurian age are also found at several points lying well within the margin of the Nucleus, as for instance in the Ottawa River about Pembroke at a distance of fifty miles, and at Lake St. John at the head of the Saguenay River at a distance of one hundred and thirty miles from its present limit. There is reason to believe that a similar outlier exists in the interior of the northern part of the Peninsula of Labrador, so that the Lower Paleozoic sea must also have covered considerable areas in the eastern half of the Protaxis, where now nothing but Laurentian is to be seen. In that portion of the Protaxis lying to the west of Hudson’s Bay strata of Cambro-Silurian and Devonian age extend up from the basin of Hudson’s Bay on the east and from the great plains on the west far over the Laurentian Plateau and probably, according to Dr. Dawson, originally inosculated. Strata of Upper Silurian and Devonian age are not known to exist in the eastern half of the Protaxis, of which the typical Laurentian area forms part, with the exception of a small outlier of Niagara age on Lake Temiscamangué at the head waters of the Ottawa—neither do any other deposits of later age occur with the exception of the Glacial Drift. What evidence there is, therefore, would rather indicate that the area, during late Paleozoic, Mesozoic and earlier Tertiary times, was out of water. If so, it must have undergone during this great lapse of ages a process of deep seated decay and denudation, culminating in the extensive glaciation to which it was subjected in Pleistocene times.

During this latter period the whole area was exposed to ice action, with the exception of the highest part of the Nucleus—the mountains of the Labrador coast—which, except toward the base, are still “softened, eroded and deeply decayed.”⁷ This extensive denudation served to remove all but mere remnants of any Paleozoic strata originally deposited on the Archean of this area, while the deep decay of the Archean rocks themselves would account for the immense numbers of gneiss boulders in the drift, which in

all probability are but smoothed cores of “bowlders of decomposition.” That an immense amount of material was removed from the surface of the area during the glacial age is shown by the immense quantities of Archean material which occurs scattered over the surface of the Nucleus itself, as well as in the drift to the south. The glaciation, with the depression and uplift which succeeded it, was the last episode in the evolution of this “original” Laurentian area and one which impressed upon it its present surface characters and type of landscape.

It is now an immense uneven plateau, comparatively slightly accentuated except along the Labrador coast. The surface is covered with glaciated hills and bosses of rock with rounded, mammilated, flowing contours interspersed with drift covered flats and studded with thousands upon thousands of lakes great and small. A country which in the far north is often bleak and desolate, but to the south, where it is covered with luxuriant forest, is often of great beauty, especially when clothed with the brilliant foliage of autumn. Even now, however, it is passing into a further stage of its history, the smooth or polished glaciated surfaces are becoming roughened by decay, the softer gneissic and limestone strata are again commencing to crumble into soil, and a new epoch has been inaugurated in which the marks of the ice age are being gradually effaced.

FRANK D. ADAMS.

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FOOTNOTES

¹ Accepting the distribution of the Laurentian in the far north, given by Dr. G. M. Dawson, as correct, the area is 2,001,250 square miles. This does not include the outlying and separated areas occurring in Newfoundland, New York State and Michigan.

² See also, The Geological History of the North Atlantic, by Sir William Dawson, Presidential Address, B. A. A. S., 1886.

- ³ See FRANK D. ADAMS—“*Ueber das Norian oder Ober-Laurentian von Canada*,” *Neues Jahrbuch für Mineralogie, etc., Beilageband VIII.*, 1893.
- ⁴ A. C. LAWSON.—“Notes on the Pre-Palaeozoic surface of the Archean Terranes of Canada.” *Bulletin of the Geological Society of America*. Vol. 1, 1890.
- ⁵ “A Fragment of Primeval Europe.” *Nature*, August 26, 1888.
- ⁶ G. M. DAWSON.—“Notes to accompany a geological map of the northern portion of the Dominion of Canada.” *Report of the Geological Survey of Canada*, 1886. p. 9, R.
- ⁷ ROBERT BELL.—“Observations on the Geology etc., of the Labrador Coast, Hudson’s Strait and Bay.” *Report of the Geological Survey of Canada*. 1882–3–4, p. 14, DD.

MELILITE-NEPHELINE-BASALT AND NEPHELINE-BASANITE FROM SOUTHERN TEXAS.

These basaltic rocks were collected by Professor Dumble and Mr. Taff, in Uvalde County, southern Texas. On the geological map of the United States, compiled by C. H. Hitchcock, 1886, there are two of the localities marked near the boundary of the Cretaceous and earlier Tertiary formation, between 99° and 100° longitude, and on the 29th degree of latitude. According to the statement of Professor Dumble, one part of the rocks appears in dikes in the upper portion of the lower Cretaceous formation, while the other forms hills and buttes. Upon microscopical examination it is evident that the specimens collected belong to two different groups of rocks. The microscope shows that those occurring in dikes consist of typical melilite-bearing nepheline-basalt, while those making up hills and buttes are nepheline-basanites tending toward phonolites in composition.

The melilite-nepheline-basalts have a typical basaltic appearance. In a dense black groundmass, the only phenocrysts seen by the naked eye are numerous olivines. Under the microscope there appear in addition to the olivine the following minerals: augite, nepheline, melilite, magnetite and perovskite. As to the proportion of nepheline and melilite, it can be said, that in nearly all the specimens examined, the two minerals are found in about the same amount. For this reason these rocks can be placed under the head of nepheline-basalt as well as under that of melilite-basalt, or they may be called melilite-nepheline-basalt. Only one of the specimens is entirely free from melilite. Feldspar is wholly wanting. All of the specimens are in a very fresh condition, and even the melilite shows only slight indications of decomposition. The specimen free from melilite corresponds in structure and composition with the other specimens, except for the absence of melilite and perovskite, and so they may be described together.

All the rocks are porphyritic, since they bear large phenocrysts of olivine. Under the microscope the olivine is colorless and transparent, and only shows indications of serpentinization along the edges and fissures. It contains rounded inclusions of glass, abundant in some sections, besides octahedrons of magnetite, and others that are transparent with a brownish violet color. Whether the latter are a mineral of the spinel-group or belong to perovskite, with which they accord in color, could not be decided.

Augite occurs in only one generation; phenocrysts of augite are wanting. In the rather coarse-grained groundmass, it becomes the most abundant constituent. The mineral shows a grayish-brown color, common to basaltic augite, sometimes with a tint of violet. It generally forms well-shaped crystals, rarely irregular grains, and bears inclusions of magnetite and glass.

Melilite occurs in the groundmass in large and well-shaped crystals, its dimensions never becoming as small as those of many of the augite crystals. They may be designated as micro-porphyritical phenocrysts. Cross sections parallel to (001) reach a diameter of 0.5 mm. The shape of the melilite is the common one, tabular parallel to (001). The diameter of the tables generally exceeds their thickness from four to six times. Sections parallel to the prism-zone, therefore, are lath-shaped and the vertical axis lies perpendicular to their length; the axis of greatest elasticity coincides with the vertical axis. Between crossed nicols these sections show the particular blue interference colors characteristic of melilite and zoisite. Sections perpendicular to the prism-zone are eight-sided by reason of the planes (110) and (100), but frequently the outlines are rounded. In some of the sections examined the melilite incloses minute opaque grains arranged zonally, which present very sharply the prismatic outlines of their host. Besides the two prismatic faces above mentioned, there is also a ditetragonal prism, the angle of which upon the adjoining faces of (110) and (100) was found to be nearly equal, 20° - 22° . According to this measurement the prism must have approximately the position of (940); the angle of the latter upon (110) is $21^{\circ} 2'$, the angle upon (100) = $23^{\circ} 58'$. A particular phenomenon in the growth of the melilite is the fact that the base does not generally present an even plane, but shows a conical depression. The shape of the lath-shaped sections then resembles the profile of a biconcave lens. Sections parallel to the base are isotropic between crossed

nicols and show, when they are not too thin, an indistinct dark cross in convergent light. The cleavage parallel to (001), the cross-fibration of the lath-shaped sections and the occurrence of the spear-shaped and peg-shaped inclusions arranged parallel to the *c* axis (the so-called *Pflockstruktur*) are very distinct. Inclusions of pyroxene, magnetite and glass are common; as already mentioned, these inclusions are generally arranged in zones. In sections parallel to (001) they fill the central parts of their host, and often make up two or three concentric zones. These sections closely resemble leucite because of their rounded shape, the arrangement of the inclusions and the lack of double refraction. Melilite becomes nearly colorless and transparent, but in comparing it with the white, colorless nepheline, it shows a feeble yellow tint. Decomposition has taken place to only a small extent; it begins along the cross-fibration, and greenish-yellow alteration-products result, the fibres of which are perpendicular to the length of the lath-shaped sections.

Nepheline is always fresh, colorless and transparent; it rarely exhibits a regular shape, but generally forms an aggregate of irregular grains, cementing the other components; it is evidently the latest formed mineral in the rock.

There is abundant magnetite besides perovskite, the common associate of melilite, which occurs in small octahedrons and irregular grains. The perovskite becomes transparent with a brownish-violet color, and shows in some sections a feeble, abnormal double refraction. There appears to be no isotropic base in the normal rock, but if any is present, it must be in a very small amount. There are coarser grained spots in the rock, which are rich in a partly chloritized base, and in which nepheline occurs in well-shaped crystals.

The second group of rocks, as already mentioned, falls under the head of nepheline-basanite poor in olivine. And since the specimens bear sanidine phenocrysts beside plagioclase, it forms a transition to phonolite. The rock-specimens have a more andesitic than basaltic appearance. Numerous phenocrysts of hornblende and augite are imbedded in the dense bluish-gray groundmass. The next most abundant mineral is nepheline in the form of phenocrysts, in part well-shaped crystals, in part rounded, the largest of which are 0.5 cm. in diameter. The nepheline differs from the feldspar in having a grayish color and greasy lustre. Phenocrysts of feldspar

and crystals of olivine are scarce. Beside these components, the rocks contain apatite, some titanite and iron ores. Under the microscope olivine is seen to be scarce. It is fresh and shows the normal properties. It contains minute octahedrons of picotite and in some sections abundant inclusions of a liquid with moving bubbles.

The amphibole mineral is a typical basaltic hornblende. It becomes transparent with a dark reddish-brown color and exhibits a strong pleochroism according to the following scheme:

a, brownish yellow, **b** and **c** dark reddish brown. Absorption, $c > b > a$.

The angle of extinction was examined in sections cut approximately parallel to the clinopinacoid (010) and was determined to be very small. This fact and the dark reddish-brown color are in all probability due to a high amount of Fe_2O_3 . The dependence of the angle of extinction upon the amount of Fe_2O_3 in minerals of the amphibole group has been recently established by Schneider and Belowsky. The basaltic hornblende shows the well-known dark borders produced by reabsorption by the magma in an early stage of consolidation. In many cases nothing of the original mineral is preserved; the whole hornblende is replaced by a fine grained aggregate of pyroxene and magnetite, presenting clearly the outlines of the absorbed mineral.

The group of pyroxenic minerals is represented by two monoclinic augites. One of them exhibits a violet-gray color in thin section and belongs to the basaltic augites; the other one becomes transparent with a dark green color. Both form numerous phenocrysts, but the first occurs somewhat more frequently. They occur as single crystals and are also grown together in a zonal manner, the green one always forming the center, the gray one the outer parts of the crystals. Hence the gray augite is the younger. The pyroxene in the groundmass shows the same color and properties. The pleochroism of the two minerals is as follows:

Gray augite.

a Brownish-yellow

b } Violet-gray

c

Green augite.

Light yellowish-green

Dark gray-green

Dark green.

The angle of extinction, $c: c$, is large and, as may be seen in the zonal crystals, it is somewhat larger in the gray pyroxene than in the green. The extinction in sections cut approximately parallel to (010) has been observed to be about 47 degrees (gray augite) and 41 degrees (green augite). The two pyroxenes show in addition to the cleavage parallel to (110) another but less distinct one parallel to (010). Inclusions of magnetite, apatite and glass are common.

Phenocrysts of feldspar are scarce. In part they show the polysynthetic twinning lamination of plagioclase; in part the latter is wanting and one of the latter feldspars, which was isolated and examined for specific gravity and optical properties, was found to be sanidine. Phenocrysts of nepheline are more frequent than those of feldspar. The mineral appears partly in the form of short-prismatic crystals, partly in rounded grains. It presents distinct cleavage, parallel to (1010) and to (0001), and the usually observed optical properties. Isolated grains are decomposed by hydrochloric acid with the separation of gelatinous silica; the resulting solution when evaporated gives numerous cubes of NaCl. Inclusions are scarce; there are fluid cavities with moving bubbles, generally arranged in rows, besides some pyroxene crystals.

Apatite forms short and stout crystals always filled with inclusions of liquids. The opaque ore grains, judging by their ready solubility, belong to magnetite. The groundmass of these rocks consists essentially of pyroxene in well-shaped prisms, lath-shaped feldspar, without twinning lamination or in single twins according to the Carlsbad law and nepheline. The feldspar of the groundmass in all probability is mostly sanidine. Nepheline is abundant and occurs in well-shaped crystals. Small patches of a colorless base occur between the crystalline components.

The structure of the rocks is hypocrystalline-porphyritic on account of the occurrence of an isotropic base and the repetition of the crystallization of pyroxene, nepheline and feldspar. Although the specimens by their whole habit and structure belong under the head of nepheline-basanite poor in olivine, the presence of sanidine as phenocrysts causes them to form a transition to the group of phonolites. Unfortunately, analyses of these rocks have not yet been made.

A microscopical examination of the basaltic rock from Pilot Knob, near Austin, Travis County, was made for the purpose of comparison with the rocks from southern Texas just described. The rock was found to be a nepheline-basalt porphyritic with numerous phenocrysts of olivine. The fine grained groundmass consists essentially of augite-crystals cemented by non-individualized nepheline in very small amount.

A. OSANN.

SOME DYNAMIC PHENOMENA SHOWN BY THE BARABOO QUARTZITE RANGES OF CENTRAL WISCONSIN.

The quartzite ranges of Baraboo extend east and west for about thirty miles, one lying north, and the other, the main range, lying south of the City of Baraboo. The geology of this district is admirably given by the late Professor Irving.⁸ Not only is the general geology clearly described, but remarkably accurate descriptions are given of the character of the quartzite, and the phenomena shown by it, considering the fact that the report was written nearly twenty years since. The unconformity existing between the quartzite and the Cambrian was later more fully described.⁹ The induration of the Baraboo quartzite has been explained as due to the enlargement of the original quartz grains; and to the deposition of independent interstitial quartz.¹⁰ The present note is based upon recent observations on the East Bluff at Devil's Lake and on the exposures at the Upper Narrows of the Baraboo River.

The section across the ranges, as given by Irving, is shown by [Fig. 1](#). The two ranges together, as thus represented, are less than the north half of a great anticline, the south side of the south range being near its crown. This structure involves a very great thickness of quartzite, and was offered with reservation by Professor Irving. He says: "The hypothesis is not altogether satisfactory. The entire disappearance of the other side of the great arch, as well as the peculiar ways in which the ranges come together at their extremities are difficult to explain by it. It may be said in this connection that the dip observations toward the west are not so satisfactory or numerous as they might be." The question naturally arises whether or not the great width of the ranges in the central part of the area may not be partly explained by monoclinical faulting, and thus reduce the supposed thickness of the beds.

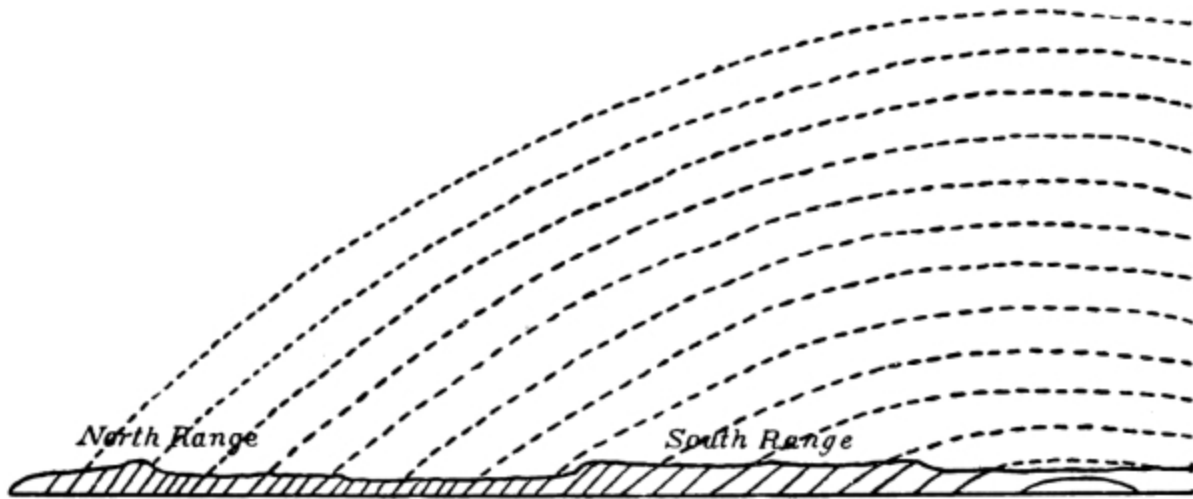


FIG. 1.—Ideal Sketch, showing structure and amount of erosion of the Baraboo Ranges.

After Irving.

Scale natural, 12,000 feet to the inch.

The layers of quartzite are ordinarily very heavy, but the changing character of the original sediment is such as to make it easy to follow the layers. Some beds were composed of fine grains of quartz, mingled with clayey material, others of coarse grains with little clayey material, and others of pebbles so large as to pass into an unmistakable conglomerate. The pebbles of the conglomerate are mainly white quartz and red jasper. It is thus easy to discriminate the bedding of the series from the heavy jointing which occurs, cutting the bedding in various directions, and from a secondary cleavage and foliation which occurs in certain localities.

From the general work of many geologists on dynamic action in folding, it is to be expected that the amount of movement necessary for accommodation between beds, and consequently the dynamic metamorphism resulting from shearing, would be less near the crown of the anticline than on the leg of the fold. That is, dynamic metamorphism ought not to be so extensive in the south range as in the north range. The facts described by Irving,¹¹ and those noted by me, fully agree with this anticipation. The central parts of the heavy, little inclined beds of the south range are largely indurated by simple enlargement. The pressure has not

been sufficient to obliterate the cores, but has apparently granulated the exterior of some of the larger fragments, as in hand specimens the exteriors of the large blue quartz grains are white. Very generally the grains show slight wavy extinction. A few of them are distinctly cracked. The crevices thus formed and those in the interstices have been filled in large part by infiltrated silica, but their positions are plainly indicated by difference in extinction, by bubbles, by iron oxide, or by secondary mica which has taken advantage of the minute crevices.

However, as described by Irving, between the heavy beds of quartzites are often layers, cut by a diagonal cleavage which dies out in passing into the thick beds. The layers showing cleavage sometimes pass into those showing the beginning of foliation, the rock then nearing a schist. In the centers of the schist zones, the schistosity approaches parallelism with the bedding, and in passing outward curves from this direction until it crosses the bedding at an angle, at the same time becoming less marked and grading into ordinary cleavage, which dies out in the quartzite. Upon the opposite side the transition is of the same character, but the curve is in the opposite direction.

Irving apparently regarded these shear zones as originally beds of a different character from the adjacent quartzite, and his conclusion is fully borne out by the thin sections. The microscope shows that the grains of quartz are of small size, and separated to a greater or a less extent by interstitial clayey material. Because of this partial separation of the grains of quartz, they have not been granulated to the extent that one would expect from the schistosity of the rock, most of the original cores being plainly visible. They, however, often show wavy extinction and even cracks, but not to a greater degree than the grains in the massive quartzite; for in the latter the full stress of the pressure has been borne by the grains in full touch, not separated by a plastic matrix, as are the grains of quartz in the argillaceous layers. In the matrix of the schist are numerous small flakes of muscovite, arranged with their longer axes in a common direction, much finely crystalline quartz, and a good deal of iron oxide.

It is concluded that the clayey character of the beds, and, consequently, the greater ease of movement within them, has located the slipping-planes and shear-zones, necessary in order to accommodate the beds to their new positions. On the south range, near Devil's Lake, these shear-zones are

generally not more than six or eight inches wide. They may be well seen just back of the Cliff House, and on the Northwestern Railway, about one-half mile south of this house. All of these shear-zones are parallel with the bedding, and illustrate the possibility, so far as I know first mentioned by H. L. Smyth, that a crystalline schist, with schistosity parallel to bedding, may be produced by shearing along the bedding-planes.

On the railroad track, near the locality where these shear-zones may be seen, is also an almost vertical shear-zone, two to four feet wide. It therefore cuts almost directly across the beds of quartzite, which here incline to the south about twelve or thirteen degrees. Throughout this band, the quartzite is broken into angular trapezoidal fragments, the longer directions of which are vertical, and which may be picked out with the hammer. In certain parts of the zone well-defined gruss or friction clay produced by the grinding of the fragments against one another, has been produced. This is clearly a plane of faulting. How much the throw of this fault is it is not easy to say, as the heavy beds of quartzite are so similar that it is impossible to certainly identify them. At this place there is, however, a change in the character of the quartzite, layers of light color being overlain by other beds, which are more heavily stained with iron oxide. This same succession is seen on both sides of the fault, and if beds of like character correspond, the amount of the throw is twenty to thirty feet, and the south side has dropped relative to the north side. In other words, the faulting is in the right direction to reduce the theoretical thickness of the sediments as given by Irving. The district has not been closely examined for other faults, but the existence of one fault, even of a minor character, suggests that a careful study of the whole area with reference to faulting should be made, in order to determine what deductions may possibly be made from Irving's estimate of the probable thickness of the quartzite.

At the upper narrows of the Baraboo, near Ablemans, we are on the north leg of the anticline. The dip is throughout from seventy to ninety to the north, and in some places the layers are slightly overturned. The slipping along the bedding has here been much greater. While in this area there are heavy beds of quartzite which have not suffered great interior movement, other beds have been sheared throughout, being transformed macroscopically into a quartz-schist, but the foliation is strongly developed. In other places, as described by Irving,¹² where the rock is a purer quartzite,

for a distance of 200 feet or more across the strike, the rocks have been shattered through and through, and re-cemented by vein quartz.

For the most part the rock is merely fractured, the quartz fragments roughly fitting one another, but there are all gradations from this phase to a belt about ten feet wide of true friction conglomerate, the fragments having been ground against one another until they have become well-rounded (a Reibungs breccia). Between the boulders of this zone is a matrix, composed mainly of smaller quartzite fragments. The whole has been re-cemented, so that now the mass is completely vitreous. This belt of friction conglomerate at first might not be discriminated from the Potsdam conglomerate, immediately adjacent, but a closer study shows how radically different they are. In one the cementing material is vein-quartz; in the other the sandstone has been feebly cemented by quartz enlargement.

A movement later than the one which produced the cemented fractured rocks and breccia has broken broad zones of the massive beds of quartzite into lozenge-shaped blocks, the longer axes of which are parallel to the bedding and movement. These later-formed blocks have not been re-cemented by secondary quartz, and the cracks are taken advantage of in quarrying, the fragments being easily picked apart. Thus the rock has been affected by at least two dynamic movements, separated by a considerable interval of time.

The shear-zones, often several feet in width, particularly affect the more finely-laminated layers, which are lean in quartz, while the relief in the more massive layers has resulted in complex fracturing. In the first phase of production of the schist, the irregular fractures pass into rather regular fractures, cutting the beds nearly at right angles. As the action becomes more intense in the more argillaceous beds, the angle of fracture, or cleavage, as it may now fairly be called, becomes more acute, and in the most intense phase this cleaved rock passes into a well-developed schist, the foliation of which is parallel to the bedding. The phenomena of shearing are here therefore very similar to those at Devil's Lake, except that the process has gone farther.

When studied in thin section, the massive beds of quartzite show more decided effects of dynamic action than at Devil's Lake. However, the major portions of the grains of quartz have distinct cores which are often

beautifully enlarged. In some cases nearly every grain has thus grown, perfectly indurating the rock. But, also, nearly every grain of quartz has a wavy extinction, and many of them have been fractured, as mentioned of a few of the quartz grains of the quartzites of the south range. In one case the pressure has been so great as to produce rather numerous roughly parallel lines of fracture. It is thus seen that the dynamic effects are not confined to the schist zones, but are also prominent within the heavy beds of quartzite. This was to be expected; for while the major part of the accommodation necessary to bend the rock mass as a whole took place along the shear zones, the accommodation required to bend each of the rigid heavy beds of quartzite must have taken place within each layer. To the consequent intense pressure and the rubbing of the grains over one another, are wholly attributed their wavy extinction and fractures.

In the schists of the shear zones, as at the south range, the thin sections show that the original quartz grains were small; interstitial material was present, and mica has developed more largely than in the quartzite. However, in the most crystalline phases, the fragmental cores of the quartz grains and their frequent enlargements are plainly seen. Thus the shearing has not been sufficient to produce a completely crystalline schist, although this would not be macroscopically discovered, unless it were suspected because the rock is not thinly foliated.

As the dip of the quartzite is so steep at this locality, it is difficult to say how far the shifting of the beds over one another lessens the apparent thickness. The shear zones as well as the friction conglomerates appear to be parallel to the bedding. If they are exactly so, this shearing action would necessitate an estimate of the original thickness greater than now shown, since the shear zones probably have less width at the present time than the beds from which they were originally produced.

Cutting the bedding are heavy joints inclined to the north at an angle of 20° to 30° . If slipping had occurred along these in the right direction, this might cause a small thickness of beds to have a great apparent thickness. However, the schists above described weather out on the face of the cliffs, and are therefore marked by recessions in the walls. If slipping parallel to the jointing had occurred since the schists were formed, these depressions ought not to match on opposite sides of the joints; but, on the contrary, they continue unbroken from foot to top, and probably the joints were formed

simultaneously with or later than the belts of schist. Consequently, at the upper narrows of the Baraboo no evidence was found of faulting which could reduce the estimated thickness of the quartzite as given by Irving.

As Irving clearly saw, bearing strongly in favor of the theory of a great fold, is the increasing steeper dip of the layers in passing north. The phenomena of movement and metamorphism corresponding so exactly to those required by a simple fold, the question may be asked if these are not evidence of some weight in favor of the general correctness of Irving's conclusion as to the structure. Had monoclinal faulting extensively occurred, it would not have been necessary to have had so great a readjustment of the beds as has been shown to occur by the schists, cleavage, and the exceedingly intricate macro-fracturing and micro-fracturing of the rock beds and their constituent particles.

In addition to the phenomena described by Irving, in summary, the Baraboo quartzite ranges show results of dynamic metamorphism as follows: A fine example of the Reibungs Breccia may be seen. A fault zone of limited throw exists. All phases are exhibited, between a massive quartzite, showing macroscopically little evidence of interior movement through a rock exhibiting in turn fracture and cleavage, to a rock which macroscopically is apparently a crystalline schist. The foliation of the schists is parallel to the original stratification, being consequent upon the movements of the beds over one another, readjustments occurring mainly in the softer layers. In thin sections the schists still give clear evidence of their fragmental origin, but also show the mechanical effects of interior movement. These same effects are apparent within the heavy beds of quartzite, some readjustment of the particles to their new positions being here also necessary. There is no evidence that the semi-crystalline character of the schist and quartzite are due to high heat. Nowhere are the particles fused. So far as they are destroyed it is by fracture, and the rock is again healed by cementation.

The rock, in its most altered condition being a semi-crystalline schist, and in other parts showing less change, can be connected with its original state. Had the folding been more intense, it is reasonable to suppose that the entire rock would have been transformed into a completely crystalline quartz-schist, showing no evidence of clastic origin, and possibly the foliation throughout would have corresponded to the original bedding.

FOOTNOTES

- ⁸ The Baraboo Quartzite Ranges, by R. D. Irving. In Vol. II, Geol. of Wis., pp. 504–519.
- ⁹ The Classification of the Early Cambrian and pre-Cambrian Formations, R. D. Irving. In 7th Annual Rep., U. S. G. S., pp. 403–408.
- ¹⁰ Enlargement of Quartz Fragments and Genesis of Quartzites, by R. D. Irving and C. R. Van Hise. In Bull. 8, U. S. G. S., pp. 33, 34.
- ¹¹ The Baraboo Quartzite Ranges, by R. D. Irving. In Vol. II, Geol. of Wis., pp. 510, 516.
- ¹² The Baraboo Quartzite Ranges, by R. D. Irving. In Vol. II, Geol. of Wis., p. 516.

THE CHEMICAL RELATION OF IRON AND MANGANESE IN SEDIMENTARY ROCKS.

Iron and manganese are frequent constituents of sedimentary rocks, in some places occurring finely disseminated through sandstones and shales, or forming a part of limestones, in other places forming the mass of the deposit in which they occur. They are both derived primarily from similar, and often from the same sources, and are in many respects alike in their chemical behavior in nature. For these reasons it is to be expected that they would frequently, if not generally, be deposited in intimate association. Such is found to be the case, and iron and manganese are often closely associated in the same deposits. Very often, however, iron and manganese deposits occur close together, but distinctly separated, while sometimes extensive deposits of iron, and less commonly of manganese, occur with little or almost no association with each other.

It is the object of the present paper to discuss the agencies which are instrumental in causing these substances to be deposited sometimes together and at other times separately. The subject is of interest as showing how slight differences in the chemical behavior of their salts may cause the almost complete separation of metals once intimately associated.

THE CONNECTION OF IRON AND MANGANESE IN NATURE.

A few words concerning the relation of manganese to iron in nature will perhaps make the following discussion clearer. One of the most common modes of occurrence of manganese is with iron, though extensive deposits containing manganese more or less free from iron often occur. When associated with iron, manganese occurs with it in various ways. Sometimes the two are intimately mixed, so that they have the appearance

of a homogeneous mass, resembling iron ore when iron is in the preponderance and manganese ore when manganese predominates. In such cases there appears to be no tendency to combine in one fixed proportion, though, as iron is a much more abundant substance than manganese, the mixture most commonly contains an excess of iron, and exists in the form of a manganiferous iron ore. The manganese, when not intimately mixed with the iron, may occur in it in pockets or as scattered nodules and concretions. Such occurrences as those described are frequent in the Lake Superior iron region, the Appalachian Valley of the eastern states, in Nova Scotia, Arkansas, Colorado, New Mexico and innumerable other places. In Virginia very common occurrences are alternating layers of iron and manganese ore. The iron in such cases is generally in the larger quantities and the more continuous deposits; while the manganese is often represented by thin lenticular layers or by bands of nodules.

From such cases, where iron predominates, there are all gradations in admixture, up to the rarer cases where manganese predominates. Frequently a given geologic horizon is characterized by both iron and manganese, though in one case it may contain only iron, in another only manganese, and in still another iron and manganese mixed in various proportions. A remarkable case of this is seen in the iron and manganese horizons immediately above, or a short distance above, the Paleozoic quartzite, on the east side of the Appalachian Valley, especially in the Valley of Virginia.¹³ Here deposits of iron ore, of manganese ore, and of both ores mixed, are found at various points along the same geologic horizons. Similar alternations also occur in the Lower Silurian novaculites of the Ouachita Mountains of Arkansas,¹⁴ in Cebolla Valley, in Gunnison county, Colorado,¹⁵ and in many other places. In many cases certain horizons are characterized over large areas by iron alone, and but little manganese, as is well seen in the Clinton formation and in the Tertiary iron-ore horizons of Arkansas and Texas; while, on the other hand, some areas of certain horizons contain considerable quantities of manganese and very little iron, as is seen in parts of the Marine limestone in New Brunswick and Nova Scotia, and also in parts of the metamorphosed Cretaceous shales of California.

THE SOURCE OF IRON AND MANGANESE IN SEDIMENTARY ROCKS.

The iron and manganese contained in sedimentary strata may be considered as derived primarily from the decay of pre-existing rocks. Some of the later sedimentary rocks may have derived a part or all of their iron from older sedimentary rocks, which, in turn, had derived their iron and manganese from still older rocks. In this way the iron and manganese in a given geologic horizon may have formed a part of various older horizons before they reached their present resting place, but, in every case, their primary source can be traced back to the original materials from which sedimentary rocks were first formed. In certain cases the sea water has supplied a certain amount of iron and manganese to sedimentary rocks, but in such cases the sea water acts only as a carrier of these materials from the land areas or from submarine sources to the strata then forming.

THE TRANSPORTATION OF IRON AND MANGANESE IN NATURE.

The process that goes on in this interchange of iron and manganese from older to younger rocks is as follows:

(1) The conversion, by surface agencies, of the minerals containing iron and manganese into forms that can be taken into solution by surface waters.

(2) The solution of the iron and manganese in surface waters, acidulated with organic and sometimes inorganic acids, and their transportation in this form from the areas of older rocks to areas over which younger rocks are being deposited.

(3) Finally, the precipitation in one or more of several ways of the iron and manganese contained in solution.

The iron and manganese thus chemically precipitated may be deposited either with mechanical sediments, such as sand, clay etc., or without them. If the deposition of mechanical sediments is largely in excess of the precipitation of iron and manganese, the final products will be beds of ferruginous shale, sandstone, etc., common in many geologic horizons. If

the precipitation of iron and manganese is in excess of the deposition of mechanical sediments, the resulting products are deposits of more or less pure iron and manganese ore. Between these two extremes there are all gradations in the admixture of the iron and manganese with mechanical sediments.

Frequently the iron and manganese which were originally finely disseminated through shale, sandstone, etc., are subsequently concentrated into bodies of comparatively pure ore, and very commonly this concentration takes place by a process of re-solution of the iron and manganese and re-deposition by replacement with limestone, or, more rarely, with some other material. The limestone or other material which thus acts as a precipitant is often in the same series of strata from which the iron and manganese were removed, and thus these two substances, which were once in a finely disseminated condition, may be converted into deposits of comparatively pure ore and yet remain in the same general series of strata in which they were originally deposited. A remarkable case of this is seen in the iron deposits of the Penokee series in Michigan and Wisconsin,¹⁶ to be mentioned again on page 370. It has also been suggested by H. D. Rogers¹⁷ that certain siderite deposits in the Coal Measures were formed by the conversion of finely disseminated sesquioxide of iron into carbonate of iron by organic matter, and the subsequent segregation of the carbonate as now found in layers and nodules.

The surface waters that carry the iron and manganese to the strata being deposited at a given time are sometimes derived from areas in which iron predominates, sometimes from areas in which iron and manganese are both abundant, and sometimes, though rarely, on account of the scarcity of such regions, from areas in which manganese largely predominates over iron. If iron and manganese were always precipitated from these waters in similar chemical forms and under the same conditions, it would be expected that the strata deriving their iron and manganese from surface waters would contain those substances in the same relative proportions as they had existed in the rocks from which they were derived, and that they would be in an intimately mixed condition. Such is doubtless often the case, or at least approximately so; but it is also often the case that iron and manganese occur in separate deposits, yet in close proximity to each other and often alternating along the same horizon. Besides this, the two substances

frequently form parts of the same deposit and yet are distinctly separate from each other. In such cases the question arises as to why the iron and manganese are not intimately mixed in the form of a manganiferous iron ore, as would be expected if they had been precipitated together. Moreover, deposits sometimes occur which are composed largely of manganese ore, with little or almost no iron, and when the source of the manganese is looked for, we often find that the rocks which probably supplied it contained both manganese and iron, and that the iron was present in a much larger proportion as regards the manganese than in the new deposit. Here again the question arises as to why the iron and manganese are not in the same relative proportions in the new deposit as they were in the rocks from which they were derived.

Four principal causes suggest themselves in explanation of this separation:

(1) It might be supposed that the deposits containing mostly iron and those containing mostly manganese received these constituents from waters derived from different sources, and carrying iron and manganese only in the proportions in which they deposited them. Under some conditions this explanation might suffice, but in many cases, such as when iron and manganese alternate along the same geologic horizon, and yet in close proximity with each other, the explanation is entirely inadequate, for the deposits are too close to each other to have been formed from different supplies of surface waters.

(2) It might be supposed that the iron or the manganese had been leached out of a deposit of the mixed ores, leaving one free from the other and depositing the dissolved ore somewhere else. This explanation, except in special cases, also appears inadequate, because the reagents in surface waters, which dissolve iron and manganese, seem to affect both about equally, so that if one were dissolved, the other should be taken up in the same way. Doubtless small differences could be found in the behavior of the organic and inorganic compounds in surface waters towards iron and manganese minerals, but they would be small as compared with the more active reactions which go on.

(3) It might be supposed that a separation could be produced by secondary concentration such as segregation, replacement, etc. This has

doubtless sometimes been the case, but where the concentrating action is not assisted by a difference in the chemical behavior of the two substances, the separation would only be on a small scale. Even in the case of concentration by replacement of limestone, if iron and manganese both acted in the same way during the replacement, it would be expected to find them deposited in an intimate mixture. Though this secondary concentration, therefore, unassisted by other agencies, would not produce all the results found in nature, yet, when it is thus assisted, it often plays an important part.

(4) The fourth, and what seems the most important, factor in the separation of iron and manganese, is that, though very often they are precipitated in the same form from the same solution, yet sometimes they are precipitated in different forms; and even when precipitated in the same form, the precipitation of one sometimes requires different conditions from the precipitation of the other. This fact will explain the alternate association and separation of iron and manganese, not only when no secondary concentration has gone on, but also in cases where such concentration has taken place, such as in the replacement of limestone, etc.

It will now be attempted to show how the various degrees of association and separation of iron and manganese found in nature may be produced by different conditions during deposition.

THE FORMS OF IRON AND MANGANESE DEPOSITED AT ORDINARY TEMPERATURES.

The mineralogical forms in which iron and manganese are deposited from solution in nature at ordinary temperatures depend on the conditions of air and water, whether of an oxidizing or a reducing nature, and on the character of the associated organic and inorganic matter either in solution or on the floor of the sea, lagoon or bog in which the deposition occurs.^{[18](#)} There are four principal methods by which iron and manganese are precipitated in nature from surface waters:

(1) By oxidation, as in the case of the precipitation of hydrous oxides and in the precipitation of the carbonate by the partial oxidation of more

complex organic salts.¹⁹

(2) By reduction, as in the precipitation of sulphide of iron by the reduction of sulphate of iron.

(3) By gaseous or soluble precipitants, as in the precipitation of sulphide of iron by the action of sulphuretted hydrogen or a soluble sulphide on a soluble salt of iron, and as in other cases to be mentioned later.

(4) By replacement of carbonate of lime or some other substance. Different forms are precipitated by these different methods.

Iron at ordinary temperatures is usually deposited from solution as the hydrous sesquioxide, the carbonate, the sulphide or the hydrous silicate of iron and potash known as glauconite. Manganese under similar conditions is deposited as the hydrous oxide²⁰ or as the carbonate, and possibly sometimes, though very rarely, as sulphide.

When solutions of organic or inorganic salts of iron and manganese are freely exposed to the action of air, as in shallow or rapidly moving streams, or in lakes and some bogs, they are quickly oxidized and both may be deposited as more or less hydrous oxides. In many bogs, however, the metals may be precipitated as hydrous oxide on the surface where oxidizing agencies predominate, but when these oxides sink and come into contact with decaying organic matter, free from the active oxidizing influences of the air, they may be reduced to carbonates.

The carbonates of iron and manganese may be precipitated when the solutions containing them are protected from oxidation by a reducing agent, such as decaying organic matter, or by being far removed from the air. Carbonate of manganese, however, is a much more stable compound than carbonate of iron, and the oxidizing conditions are often sufficiently strong to cause the deposition of iron as hydrous sesquioxide and not strong enough to change the manganese from its carbonate form. It is not uncommon, therefore, to have iron deposited in one place as hydrous sesquioxide, and manganese carried further on and deposited as carbonate, or even under special conditions deposited as carbonate with the hydrous sesquioxide of iron. Fresenius²¹ has shown that the warm springs of Wiesbaden, which contain iron and manganese among their other mineral

constituents, deposit iron in the form of hydrous sesquioxide, while manganese is carried on further in solution and deposited as carbonate. In this behavior, therefore, we have the first striking difference in the deposition of iron and manganese, and it will be further discussed later on.

The sulphides of iron and manganese differ very much in their nature and mode of occurrence. Iron is frequently deposited as sulphide, but manganese rarely occurs in that form, and when it does it is always in very small quantities. Iron forms several sulphides in nature: pyrite (FeS_2), marcasite (FeS_2),²² pyrrhotite ($\text{Fe}_{11}\text{S}_{12}$), troilite (FeS) and numerous other more complex compounds unnecessary to enumerate here. Pyrite is the commonest form of iron sulphide, and occurs in rocks of all ages, from Archean to Recent. It is formed in nature by the action of soluble sulphides or sulphuretted hydrogen on soluble salts of iron, and also by the reduction of sulphate of iron by organic matter or other reducing agents. Manganese forms two²³ sulphides, alabandite (MnS) and hauerite (MnS_2). Both minerals are very rare, and so unstable that they rapidly oxidize on exposure. Alabandite is the less rare form, and usually occurs as a subordinate constituent of certain metalliferous veins or allied deposits.

Though the sulphides of manganese are easily oxidized, they are not so unstable that, had they ever been formed in considerable quantities in sedimentary deposits, they would, even at considerable depths, have left no trace of their former presence. Moreover, the sulphide of manganese, as produced artificially,²⁴ is soluble in certain organic acids, notably acetic, and, as the conditions for the deposition of sulphides of metals in sedimentary deposits generally require the presence of organic matter, it is not improbable that some of the acids given off by such matter would be capable of dissolving sulphide of manganese. Here, then, is one reason why manganese might not be deposited as sulphide under some conditions which would cause the precipitation of sulphide of iron. Moreover, the artificial formation of sulphide of manganese (alabandite) in the laboratory is brought about most easily at high temperatures. It has also been noted that when manganese, in the form of the alloys spiegeleisen and ferro-manganese, is added to molten steel, it bodily removes a part of the sulphur; and it is thought by some metallurgists, that sulphide of manganese is formed and carried into the slag.

These and other indications of the more easy transition of manganese into the form of sulphide at high rather than at low temperatures afford another cause which might prevent sulphide of manganese from being formed in sedimentary deposits, for such deposits are usually laid down at ordinary temperatures. On the other hand, they also afford a cause which might lead to the deposition of the sulphide of manganese in certain metalliferous veins and other deposits, where the temperature at the time of deposition may have been high.

In many of the silver and lead deposits of the Rocky Mountains manganese oxides occur with the superficial oxidation products of the sulphides of other metals, and it has often been suggested that the manganese also was originally in the form of sulphide. This may be true in some cases, for alabandite has been found in a few metalliferous deposits in Colorado, Mexico, Germany, Peru and elsewhere, but in most cases, at least in the Rocky Mountains, when the level is reached at which the oxidized forms of lead, zinc, iron and other metals pass into sulphides, the manganese passes into carbonate or silicate, and remains in one or both of those forms to all depths that have been reached.

In the deposition of iron and manganese as sulphide, therefore, there is a most marked difference of behavior, and here again is a good cause for the separation of the two substances in sedimentary rocks, as will be more fully explained below.

Iron is often deposited in sedimentary formations as the hydrous silicate of iron and potash known as glauconite, and composes the mass of the large greensand beds common in Cretaceous and Tertiary strata; but manganese is not found in an exactly similar condition.²⁵ Here again, therefore, is an important difference in the modes of deposition of iron and manganese, which also will be mentioned again.

It will thus be seen that while some of the forms in which iron and manganese are deposited are the same, others differ very widely, and even similar forms are often deposited under different conditions. It is doubtless to these various forms and conditions of deposition that the alternate association and separation of iron and manganese in nature are due.

CAUSES OF THE ASSOCIATION OF IRON AND MANGANESE.

The very frequent intimate association of iron and manganese in sedimentary rocks is what would be expected from a deposition as oxide or carbonate in basins such as coastal lagoons or bogs, where the waters moved very slowly, or not at all, for under such conditions, they are often deposited together.²⁶ Moreover, it is a well-known fact that isomorphous substances have a strong tendency to combine in a homogeneous mass, and to crystallize together in different proportions. Carbonate of iron and of manganese are isomorphous with each other, and this is hence a possible cause of the frequent intimacy of their association, such as is seen in almost all manganiferous spathic iron ores, whether these ores are formed by direct precipitation or by replacement of carbonate of lime. The oxidation of such a mixture would give the common form of an intimately combined iron and manganese ore.

Since there is usually more iron than manganese in the rocks from which both metals were originally derived, the surface waters draining from areas of such rocks usually contain the metals in a similar proportion. Hence, in cases where the deposition of the carbonates of both occurs at the same spot, the isomorphous carbonates derived from the solutions have a larger percentage of carbonate of iron than of carbonate of manganese, and the resulting oxides contain the two metals in the same proportion, thus giving rise to the common low-manganese iron ores.

The hydrous oxides of iron and manganese, however, are not isomorphous,²⁷ and, therefore, when they are precipitated together, as in bog-deposits, the association is often much less intimate than in the cases just mentioned, and is simply due to the fact that, under certain conditions, the oxides of both metals are precipitated in the same place.

CAUSES OF THE SEPARATION OF IRON AND MANGANESE.

When iron and manganese ores occur in more or less separate deposits, it becomes necessary to suppose the action of influences different from those which cause the deposition of both together, and such influences are to be found in the different modes of precipitation, under certain conditions, of the two metals. It has been shown by Fresenius²⁸ that certain warm springs, on reaching the surface, first deposit hydrous sesquioxide of iron, and farther on carbonate of manganese. This not only points to the well-known fact that carbonate of iron is more easily oxidized than carbonate of manganese, but it also leads to the belief that the bicarbonate or other salt of iron in the water is more easily oxidized than the manganese salt.

An action somewhat similar to that described by Fresenius readily explains the occurrence of manganese sometimes in entirely separate deposits, sometimes in distinct but closely alternating deposits.²⁹ Under certain conditions, if the waters from which the precipitation took place were moving, the iron and manganese, owing to the difference in oxidability, as stated above, would be laid down in different places, resulting in the formation of deposits of iron ore free from manganese, and manganese ore free from iron in different positions along the plane of the same geologic horizon. Such occurrences are often seen in the iron regions of the Appalachian Valley, where there are often found, in different places along the same belt, deposits of iron ore and deposits of manganese ore in positions similar with relation to the enclosing rocks.

These conditions of moving water might also cause the occurrence of the two ores in interstratified layers, as is sometimes the case. Such a condition would result if iron were deposited in a certain place at one time, and if, later, on account of some increased facility for oxidation, iron was deposited before it reached that place, and the manganese, being less easily precipitated, were carried on and laid down upon the first deposit of iron.

Suppose the metalliferous solutions to be confined in a shallow basin, or, at least, to pass through it so slowly that they become thoroughly oxidized. Under such conditions the deposition of iron and manganese would go on continuously, and so nearly on the same spot that a comparatively homogeneous manganiferous iron ore would be formed. If the supply of metalliferous solutions were not continuous, but were intermittent, as is sometimes the case in local basins, such as coastal

lagoons, which are often dependent for their supply of water on the changes of season and the sudden fluctuations of weather, then interstratified layers of iron and manganese ore might be produced. The iron, becoming oxidized on the surface, sinks to the bottom, possibly in some cases to be converted there to the simple carbonate by organic matter. Further oxidation precipitates hydrous oxide or carbonate of manganese on top of the iron. A renewed supply of surface waters brings more solutions of iron and manganese, or else the evaporation of the water in the closed basin concentrates the materials which have not yet been precipitated. In either case there is a further alternate deposition of the two ores. [30](#)

Another process of separation of iron and manganese in nature might take place by the formation of sulphide of iron. It has already been shown that iron is sometimes deposited as sulphide and later oxidized in the same manner as the carbonate. Manganese, on the other hand, is rarely found as sulphide, and there is reason to think that the sulphide never represented the original form of any large sedimentary deposits of manganese ore (see pages 364 to 365). It seems probable, therefore, that from a solution of iron and manganese in surface waters the iron might, where the conditions are favorable, be precipitated as sulphide (FeS_2) and the manganese might be carried on in solution to be deposited somewhere else as oxide or carbonate. Subsequently the oxidation of the ores would give rise to oxide of iron from the sulphide and oxide of manganese from the carbonate; and the two ores, though occurring at the same horizon, would be separated by a greater or less distance.

After the deposition of the sulphide of iron, the conditions might change and permit the deposition, in the same place, of the carbonates of iron and manganese together. This is an easy case to imagine, and where such a deposit was exposed to surface influences, the resulting product would be oxide of iron from the underlying sulphide and a mangiferous iron oxide from the overlying isomorphous carbonates. Hence another possible cause of the frequent association of pure iron ores and mangiferous iron ores. It is possible also that after the solution of iron and manganese had been freed from the former by precipitation as sulphide, the manganese might be carried on and laid down as carbonate on a previous deposit of iron sulphide, and when such a combination was

oxidized, the result would be oxide of iron and oxide of manganese in beds closely associated but yet distinct.

By supposing the iron sometimes to be deposited in sea water as glauconite, a manner in which manganese is not laid down (see page 365), a further means of separation of the two metals would result.

Thus by alternating the conditions of the deposition of iron and manganese in different forms, a great variety of methods of association and separation of the two metals can be produced.

The above discussion refers not only to the deposits of iron and manganese ores of notable size, but also to the iron and manganese frequently found disseminated through shales, sandstones etc. In these rocks they usually form a small but often a very important part, for in many cases the iron and manganese is taken into solution from the rocks and redeposited by a process of replacement with carbonate of lime in neighboring beds of limestone, or more rarely by replacement with other rocks, thus giving rise to important ore deposits. The question of the association and separation of the iron and manganese in these replacement deposits depends on a number of conditions, the principal of which are, just as in the class of deposits that has been discussed, the conditions during deposition and the forms in which the iron and manganese are precipitated. The processes by which association and separation occur in replacement deposits differ somewhat in detail from the processes just discussed, but are based on the same principles.

Many of the iron and manganese deposits of the Appalachian region are supposed by many to be replacement deposits. N. S. Shaler³¹ in 1877 suggested that some of the iron deposits of Kentucky and Ohio were formed by the solution of iron from certain rocks, and its deposition in the form of carbonates by replacement with underlying limestone. Subsequently it was changed by oxidation to brown hematite. A notable case of replacement has also been shown by R. D. Irving and C. R. Van Hise³² in the iron deposits of the Penokee series of Michigan and Wisconsin. Here the ore is supposed to be partly a replacement of chert in a trough between quartzite and igneous rocks. The solution that contained the iron was derived from strata in the same series of rocks in which the iron was re-deposited and contained a certain amount of manganese. It is shown how the iron and manganese

were more or less separated in the replacement process and that the separation was due to the difference in the oxidability of the carbonates as explained on page 363.

R. A. F. PENROSE, JR.

FOOTNOTES

- ¹³ The exact age of the iron and manganese deposits here referred to is, in some cases, a little uncertain. Some may be Cambrian, others Silurian, but the exact determination of the age of the horizon is not a part of the present discussion. The matter has been discussed by the writer in Geological Survey of Arkansas, 1890, Vol. I., pp. 376–379.
- ¹⁴ See Geological Survey of Arkansas, 1890, Vol. I., pp. 320–325.
- ¹⁵ See Geological Survey of Arkansas, 1890, Vol. I., pp. 456–457.
- ¹⁶ R. D. Irving and C. R. Van Hise, U. S. Geol. Survey, Tenth Ann. Report, 1888–1889, Vol. I, pp. 409–422.
- ¹⁷ Geol. Survey of Penn., Vol. II, 1858, p. 739.
- ¹⁸ The solutions may be precipitated, as already shown, either with or without admixture with mechanical sediments; and there are in nature all gradations from almost pure deposits of iron and manganese ore to beds of shale, sandstone, etc. stained with iron or manganese. Subsequent concentration frequently causes decided changes in the latter deposits (see p. 370).
- ¹⁹ It has been suggested by A. A. Julien (Proceed. Amer. Assoc. Adv. Sci., Vol. XXVIII., 1879, p. 356) that in some cases the carbonates of iron and manganese may be only the fixed

residue of organic compounds of more complex form once in solution in surface waters.

²⁰ This oxide is generally in the form of the peroxide or the sesquioxide in a more or less hydrous condition.

²¹ Jahrb. des Vereins f. Naturkunde in Herz. Nassau, Vol. VI., p. 160 (Bischof).

²² Marcasite has the same composition as pyrite, but differs in crystalline form.

²³ Manganese also occurs in the mineral youngite, which contains lead, zinc, iron, manganese and sulphur, but the mineral is considered of doubtful homogeneity. (See System of Mineralogy, E. S. Dana, 1892).

²⁴ When manganese is precipitated artificially as sulphide it is usually in the form of the monosulphide (MnS), in either a hydrous or an anhydrous form.

²⁵ Manganese occurs in various hydro-silicates, but they do not appear to be deposited as sedimentary strata in the same manner as glauconite.

²⁶ If the water moved very slowly, the deposition would probably take place approximately in the same spot; if the waters moved more rapidly, the iron might be deposited in one place and the carbonate in another, in the way explained on page 363.

²⁷ The hydrous oxides of iron are not crystalline.

²⁸ See p. 363.

²⁹ Bischof suggests that the action described by Fresenius causes the separate deposition of iron and manganese; and also that it explains the occurrence of large deposits of manganese ore in regions where the iron ore contains least of that ingredient.

(See Elements of Chemistry and Phys. Geol., Vol. III., pp. 531–532.)

³⁰ In some cases these iron and manganese deposits are undoubtedly formed by the replacement of limestone or other rocks, as is further discussed on pages — to —.

³¹ Kentucky Geol. Survey, Report of Progress, Vol. III., New Series, 1877, p. 164.

³² U. S. Geol. Survey, Tenth Annual Report, 1888–1889, Vol. I., pp. 409–422.

SOME RIVERS OF CONNECTICUT.³³

Outline.—Introduction.—Topography of Connecticut: The upland plateau, its origin, date, elevation, valleys sunk beneath its surface.—Lowland on the Triassic area.—Later oscillations.—Résumé of the topography.—Early drainage.—Re-adjusted streams.—Revived streams.—Unconformable rivers, consequent or superimposed.—Pleistocene changes; the Farmington, Quinnipiac, Scantic.—Abandoned gaps.

Introduction. In order to study intelligently the history of a river, one must first become acquainted with the present physical geography of the region in which the river lies, and know the stages of its development. Therefore, before classifying the rivers of Connecticut, I shall consider the topography of the state, and in a few paragraphs outline the successive cycles in the history of its growth. The scope of this article will not permit a discussion or even a full statement of the evidence on which these conclusions are based. They have been stated at considerable length by Professor W. M. Davis,³⁴ and the reader is referred to his papers for the complete discussion. His conclusions in respect to the physical geography are accepted here without question, and form the basis for the discussion on the rivers of the state.

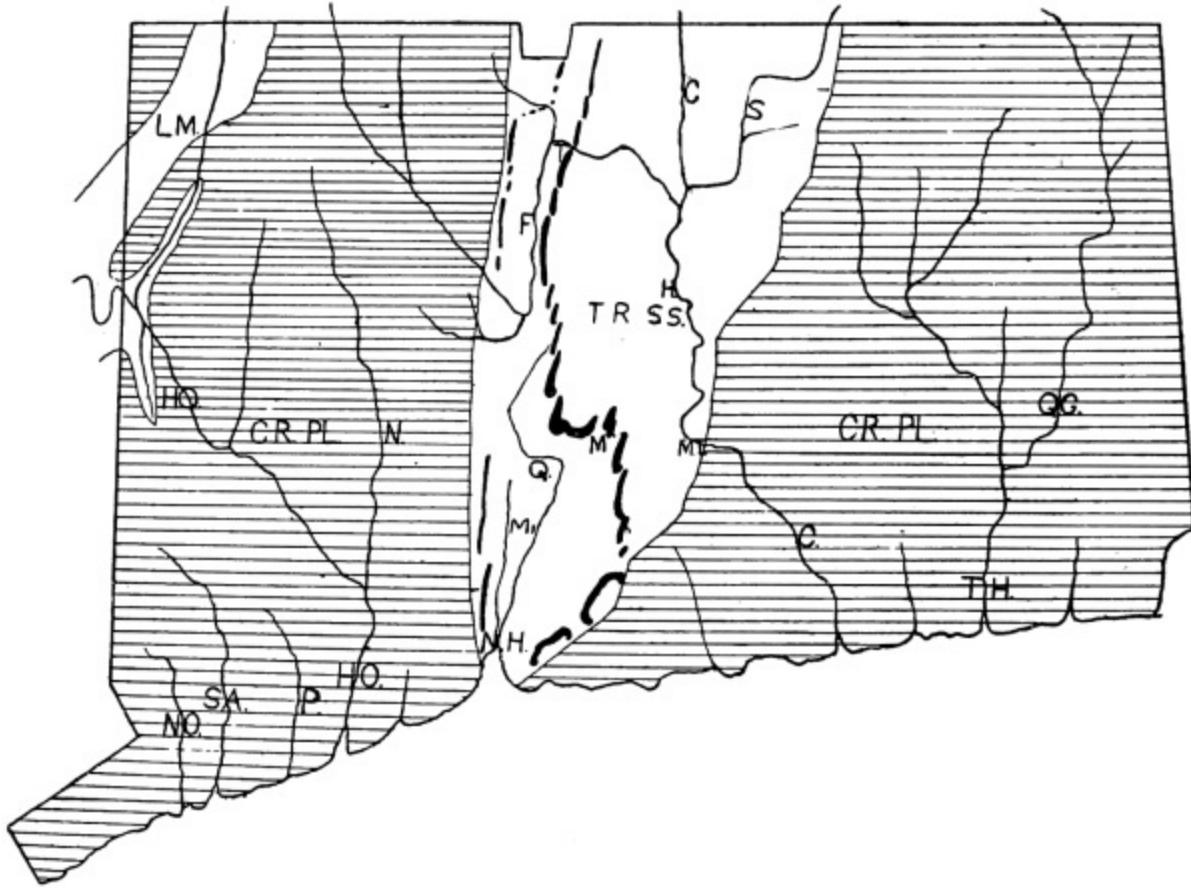


FIG 2³⁵

Topography of Connecticut. Connecticut can be said to consist of two great areas quite distinct in topography and geologic structure.³⁵ On the east and on the west are the crystalline uplands which rise from sea level along the Sound to 1,700 and 1,800 feet in the northwestern part of the state, and to 600 and 700 feet in the northeastern. These uplands consist chiefly of gneiss and granite, probably of pre-Paleozoic age, which are now much folded, faulted and crumpled. Between these two areas of crystallines is a lowland belt of Triassic sandstone and shale, twenty to twenty-five miles wide, extending from New Haven north through the center of the state and including in its borders New Haven, Meriden, Hartford, New Britain and many towns of lesser note. These sandstones form a monocline with an eastward dip of 10° to 30° , and in addition to being tilted they have been faulted since their deposition in a shallow, slowly-subsiding trough of crystallines. Their thickness is variously estimated—3,000 to 5,000 feet,

Dana; 10,000 or more, Davis. This lowland is interrupted by a series of trap ridges, which in general present steep faces toward the west, whereas their eastward slope is gradual, less than the dip of the sandstones.

The upland plateau. Suppose we ascend the highest point of these trap ridges, the old tower on Talcott Mt., nine miles west of Hartford; we are 900 feet above the sea level and more than 600 above the plain at our feet. A few miles to the west across the sandstone valley, rise the crystalline uplands, which extend far to the north and to the south. On the east across the Connecticut we see the eastern uplands. The first impression, which comes to one as he gazes upon these uplands and which is strengthened with each view, is that few hills rise above the general level of the plateau; the crest line is nearly horizontal, declining gently to Long Island Sound. Above this general level are a few rounded domes, but no sharp, towering peaks. Below it valleys have been cut, but they do not destroy the plateau-like appearance. A view from the western plateau across the sandstone valley shows the remarkably even crest line of the trap ridges, a crest line which approximates in height the uplands on the east and west. A nearer view of the upland corroborates our first impressions of the gently rolling character of the inter-stream surfaces, but we have a better view of the valleys which have been sunk beneath the general level and of the low rounded hills which rise above it. In popular parlance the country is “hilly.” It is uneven, not because there are high hills, but rather because there are deep valleys. If in imagination we fill up these valleys and the wide Triassic lowland to the general level of the broad inter-stream surfaces, we shall have constructed a gently undulating plateau, dipping to the south and east—a peneplain.³⁶

Origin of the peneplain. This is not a constructional surface, for the rocks are greatly tilted, folded and faulted, so that the surface consequent upon such disturbance must have been complex and mountainous. Long subaërial denudation upon a folded and faulted mass when the land stood much lower than at present produced this plateau. Evidently it could be produced by denudation only at or near baselevel, for the effect of erosion upon a mass high above baselevel is to accentuate its topographic relief, not to reduce it. We naturally ask ourselves, “At what stage in geologic history did this denudation occur?”

Date of the peneplain. The erosion which accomplished this great work must have commenced after the formation and dislocation of the Triassic beds, for the even crest line of the trap ridges, a part of which—perhaps all—were contemporaneous with the sandstones, is a part of the dissected peneplain; but to fix the date of the completion of the peneplain, we must turn to evidence presented in New Jersey.³⁷ There we learn that by the close of Cretaceous times, the country was eroded nearly to baselevel, and we may therefore speak of the relative position of the land and sea, to which the land was at this time reduced, as the Cretaceous baselevel, and this land surface as the Cretaceous peneplain.

Elevation of the peneplain. In post-Cretaceous, presumably early Tertiary³⁸ times, the land was elevated to nearly its present height and remained at that altitude, so far as topographic evidence shows, during Tertiary times. The proofs of this elevation are the valleys which the streams have sunk below the general level. That this was not a simple uplift, but was accompanied with tilting and warping, is clear from the following considerations. The depth to which a stream can cut its valley depends directly upon its height above baselevel. If the present surface were a peneplain uniformly elevated, the head waters and middle courses of a river would not be cut so deep in the surrounding plain as its lower course. But the reverse is true of the rivers of Connecticut. The depth of the valley increases inland, being greater in those regions where the peneplain was raised the highest. A comparison of the upper and lower valleys of the Housatonic, Naugatuck, Quinnebaug, and of the Connecticut at Middletown, where it enters the plateau, and at its mouth, will give some idea of the amount of the warping. It will not give an *exact measure* of it for several reasons: first, the upper courses of the rivers have not yet reached the present baselevel; second, the present altitude of the uplands is the result of the post-Cretaceous uplift and warping, plus a probable later post-Tertiary uplift (to be mentioned later), besides several minor oscillations, the last of which was downward, and is recorded near the coast in the drowned condition of the rivers. As has been already said, the peneplain is highest in the northwest, and gradually declines to sea level toward the south and east.

Consequences of the uplift. The consequences of this uplift are seen in the valleys, which are cut into the peneplain, and which have destroyed the

level character of the country. In the hard crystalline rocks the valleys are generally narrow and deep, with bold slopes;³⁹ where they are cut in the crystalline limestone, they are wider and more open. In marked contrast, however, is the lowland on the Triassic area in which only the trap ridges remain to tell of the former altitude of the general surface, and the immense amount of erosion which has taken place on the soft sandstones and shales. Indeed erosion has progressed so rapidly on these soft rocks, that they have been worn down almost to a new baselevel in the same length of time in which the hard crystallines have been only trenched. This fact cannot be too strongly emphasized. The broad sandstone lowland from New Haven north into Massachusetts has been carved out of the uplifted peneplain in soft rocks, during the same time in which the Connecticut has excavated its gorge in the crystallines below Middletown, and the Housatonic has opened its upland valley on the limestones. The difference in results is due not to a difference of time, but to the difference in the relative hardness of the rocks.

On the basis of this principle the age of certain river gorges to which reference will be made later can be fixed. The *narrow* passage of the Quinnipiac through a sandstone ridge southwest of Meriden cannot belong to the same cycle of erosion as the *broad sandstone lowland* on either side of it, but manifestly must be much younger. So, also, the narrow passage of the Farmington at Tariffville, where it crosses the trap ridge through a gorge free from drift, is of much later date than the *broader* valley more or less encumbered with drift which the upper part of the same river has cut in the hard crystalline schists. Cook's Gap in the trap sheet west of New Britain is much broader than either of the above, and belongs to the Tertiary cycle of erosion, although as I shall endeavor to show later, it was probably not occupied by a stream during the whole cycle. In marked contrast, also, with the Tariffville gorge is the gap by which the Westfield river in Massachusetts cuts the trap ridge. This gap was formerly broad and open—the result of Tertiary erosion—but is now filled with drift, in which the river is at present working. Since these two rivers are essentially the same in size, are now at the same level, and the rock is the same in both cases, the only explanation for the difference in the two passages is that they belong to different cycles.

To recapitulate, the results of the post-Cretaceous uplift are seen in the valleys which have been cut in the peneplain. The narrow valleys in the

gneisses and schists, the upland valleys in the limestones, the wide open, drift encumbered gaps in the trap ridge,—Cook's and the Westfield river gaps,—the broad open lowland on the sandstones, are all the result of erosion in this cycle. The Quinnipiac gorge in the sandstone, and the Tariffville gorge in the trap are just as surely of a later date. They do not at all accord with the work of the earlier cycle either in size, angle of slope, or depth.

This conclusion is somewhat at variance with an opinion expressed by Professor J. D. Dana,⁴⁰ but it seems justifiable in view of the successive cycles in the physical development of the region. In another part of this article I shall consider these gaps again in connection with their river histories, and shall give additional reasons why I venture to differ from so eminent an authority.

Length of this cycle. This cycle of erosion beginning with the post-Cretaceous uplift was not so long as the preceding cycle. In the earlier one the whole state was reduced to a peneplain; in the later cycle only the soft Triassic sandstones were brought near to baselevel. It probably lasted through Tertiary times, and was brought to a close by a slight uplift. The result of this uplift is well shown in Pennsylvania⁴¹ and New Jersey.⁴² It is not well shown in Connecticut, but there seem to be some traces of it in the trenches the rivers have cut below the level of the sandstone peneplain. However, these trenches are so much obscured by drift that a positive statement is not warranted. It may, however, be spoken of provisionally as the post-Tertiary uplift. There may have been later oscillations of small amount, probably were; here and there are shreds of evidence which point to such oscillations, but only one movement has had an effect upon the topography, which can be recognized. The fjorded condition of all the rivers along the Sound—the Norwalk, Saugatuck, New Haven bay, Niantic and Thames are the best examples—shows that within comparatively recent time there has been a slight subsidence of the land. But this movement is not to be compared in amount with those of the earlier cycles.

The drift. Over all the state in varying thickness lies the glacial drift, either in its typical unmodified development as till, or in its modified form, as river terraces, kames, eskers and sand-plains. It is of importance in this connection only as it has affected the topography of the country and so

modified the drainage. Examples of these modifications will be mentioned later.

Résumé. There was first a long cycle of denudation in pre-Triassic times, during which the contorted crystallines were worn down to a comparative level; second, a cycle of subsidence, deposition and volcanic outburst, during which the sea entered the crystalline trough, and the Triassic conglomerates, sandstones and shales were deposited with the intercalated layers of lava; third, a long cycle of elevation, folding, faulting and erosion, during which the sedimentary beds were elevated—tilted into the present faulted monocline, and this constructional surface worn down to a baselevel of erosion in late Cretaceous times. Each of these cycles probably represents the sum total of several subordinate cycles. There was, fourth, a post-Cretaceous uplift inaugurating a period of erosion lasting through Tertiary times and resulting in the formation of valleys in the hardest rocks, and a lowland approaching baselevel on the Triassic sandstones and shales; fifth, a probable late or post-Tertiary uplift, when the valleys were deepened and the lowlands trenched—obscure in Connecticut, but well shown farther south; sixth, the land, near the coast at least, is now slightly lower than it has been in the not remote past, as is shown by the fjords.

With the changes of the physical geography clearly in mind, the rivers of Connecticut may now be examined in respect to their conditions of origin, the number of cycles through which they have lived, and the approach they have made to mature old age. But at the very outset a serious difficulty is encountered, for the geological structure of the state is nowhere well described, nor have topographic maps of all the districts yet been issued. Since the structural details are to some extent unknown it is unwise in many cases to attempt more than tentative conclusions. Several of the problems to be presented cannot be considered as settled. Considerable progress toward a final settlement will have been made, however, if the conditions of the problems are made clear, various hypotheses suggested, and the attention of workers in this field called to these questions.

Early drainage. Of the drainage of Connecticut during Jurassic and Cretaceous times very little can be said. It is not even known whether it was consequent upon the Jurassic tilting and faulting, or whether these deformations were so slow in their movement that the rivers persisted in

spite of them. It may have been that the larger rivers were victorious, while the smaller were conquered and compelled to assume new consequent courses. Whatever was their origin there must have been abundant opportunities during the long erosion which resulted in the Cretaceous baselevel, and again in the period of revived and quickened degradation succeeding the post-Cretaceous uplift, for the streams to adjust themselves in a large degree to the geological structure. The contrast of hard and soft beds and the great elevation must have been potent factors in bringing to pass such a result. We expect to find the streams so far re-adjusted as to render improbable the discovery of their manner of origin.

The Housatonic, a re-adjusted stream. The best example of re-adjustment is found in the northwestern part of the state where the Housatonic and some of its branches follow well adjusted courses. From its headwaters, near Pittsfield, Mass., to New Milford, Conn., it has nearly all the way chosen its course along the Cambrian crystalline limestones in preference to the harder granites and gneisses on either side. The stratigraphical relationships of the limestone are not fully understood, but they seem to be deeply eroded anticlines and synclines, whose axes plunge north or south at various angles. The course of the river, if the drainage was consequent, was at first along the synclinal valleys, passing from one to another across the lowest points in the anticlinal ridge between them. But by a series of changes⁴³, resulting from the differential rates of erosion as hard or soft beds became exposed, the river previously to the Cretaceous baseleveling, seems to have re-adjusted its course to the softer limestones. However, there are several places where this conformity to structure does not seem to be the law; where the river departs from a limestone valley to flow for a time in the crystallines, only to return to the limestone again. The most marked instance of this is in the towns of Sharon and Cornwall, where the river leaves the limestone valley, which continues to the southwest, and flows for ten miles in a narrow gorge in the gneiss, only to again enter at its northern end a long narrow bed of limestone. The following seems to be the probable explanation. When the land stood at the elevation represented by the Cretaceous peneplain, these hard beds were below or but very slightly above baselevel, and were therefore undiscovered by the stream or had just begun to make themselves known late in the cycle. Had they been reached early in the cycle, when the stream was far above baselevel and presumably

before many of its tributaries had been developed, and when it was therefore a smaller river, it is quite probable that further re-adjustments would have occurred, and the stream been led away from the hard rocks onto the softer beds to the west; but when they were reached the stream had cut so deeply and so nearly to baselevel that it was safe from capture. After the elevation of the peneplain the stream was revived and disclosed more and more of these hard beds, but was then, owing to the development and head-water growth of its tributaries, too important a river to be diverted by any rival. A river of this kind may be said to be “conformably superimposed” in distinction to one which is superimposed from an unconformable cover.

Revived streams. It is important to recognize the effect of the post-Cretaceous uplift upon the rivers at that time established. As the land was baseleveled and the velocity of the streams decreased, they lost in large degree their cutting power and sluggishly meandered more or less in broad flood-plains. During and for a period after the uplift, their cutting power was restored to them by virtue of their increased velocity and they excavated the deep narrow valleys which we find in the crystalline highlands. The upper course of the Housatonic is a good example of a river re-adjusted to the structure during one cycle, revived by uplift to a second cycle of erosion, and in places “conformably superimposed” upon structures from which it would have been led away in the ordinary course of re-adjustment. Its tributaries, the East Aspetuck, Still, Shepaug, and Pomeraug follow courses re-adjusted in one cycle and revived in a later uplift.

We can assert with the more confidence that such was the history of the upper Housatonic, because we find in other states, in regions whose history has been the same, similar examples of “conformably superimposed” and “revived” streams. The Musconetcong and Pequest, highland rivers of New Jersey, are streams “revived” from mature old age to vigorous youth and “conformably superimposed” upon saddles of gneiss between two limestone valleys.⁴⁴

Unconformable rivers. In considering the course of the lower Housatonic we meet with some difficulty at the outset. In the southern part of the town of New Milford the river leaves the limestone belt which

continues with some slight interruptions to the Hudson, and swings sharply into the crystalline plateau in a southeasterly course until it is joined by the Naugatuck, when their united waters flow south for a few miles to the sound. The course of the lower Connecticut is even more surprising. At Middletown it leaves the broad open Triassic sandstone lowland, and through a gorge enters the plateau, which has an average elevation of 600 to 700 feet. In this plateau of crystallines the river has sunk its valley nearly to sea-level. The slopes are steep compared to the lines in the sandstone lowland, and the contrast between the two parts of the river is one of the striking features of Connecticut scenery. Several theories may be framed to account for the curious behavior of these two rivers, but none of them are free from all difficulty.

As a consequent river. The lower Connecticut has been thought⁴⁵ to be a revived river, whose course was consequent upon the post-Triassic tilting and faulting. The faulted monocline seems to have had the shape of a half-boat, ends to the north and south, and one gunwale rising toward the west, the combined effect of the tilting and faulting being to swing the river to the southeast, where the keel of the boat was lowest. The probable existence of faults, with upthrow on the east, along the eastern margin of the Triassic rocks, is a difficulty in the way of the complete acceptance of this theory. Unfortunately too little is known about the structure of the western plateau to say whether the course of the lower Housatonic could be accounted for on such an hypothesis. On this theory the Connecticut would be consequent upon the Jurassic deformation, and revived by the post-Cretaceous uplift.

It may be suggested that the southeast courses are due to the tilting of the peneplain at the time of elevation, the plateau now being, as we have seen, much higher in the northwestern part of the state than elsewhere. But the acceptance of this theory necessitates a degree of smoothness and absence of even mild relief in the peneplain, which is hardly possible. The present average slope of the plateau is but a few feet per mile, and it seems incredible that so gentle a tilting could force rivers as large as these to take new courses. Besides, if the Housatonic and Connecticut were deflected, why were not the smaller streams—the Naugatuck and Quinnebaug—also given southeastern deflections? Clearly, this explanation is not the correct one.

Superimposition. It has been suggested that these courses may be inherited from a Cretaceous cover, which formerly stretched over Connecticut for a considerable distance, but of which no traces now remain in the state. On parts of Long Island the Cretaceous deposits are found, and it is not inherently impossible nor improbable that they once stretched far over the main land. In New Jersey⁴⁶ several lines of evidence seem to show that the Cretaceous beds formerly extended across the Triassic, probably to the margin of the highland plateau. The curious drainage of the Watchung Crescent is one evidence of this, but the other proofs are along entirely different lines, so that there is apparently good evidence that the Cretaceous beds extended twenty-five miles or more farther inland. If, in the time which has elapsed since the deposition of these beds, there has been erosion sufficient to strip them off from such a broad area in New Jersey, may they not, in Connecticut, under presumably similar conditions, have been equally eroded?

There is much which makes this hypothesis attractive, and, as the facts were first studied, it seemed the most likely one. It affords a good explanation, not only for the courses of the Housatonic and Connecticut, but also for other rivers along the sound. It seems, also, at first thought, to be well supported by analogy from New Jersey. But a closer study of the situation in that state reveals marked differences in the attendant circumstances. There the soft Triassic sandstone must have been worn down to a lowland early in the Cretaceous cycle, perhaps by the close of Jurassic time or thereabouts, while the harder crystallines retained a strong relief. The slight subsidence, which marked the beginning of marine Cretaceous in New Jersey, allowed the Cretaceous sea to transgress rapidly the baseleveled sandstones to the foot of the crystalline hills, but not to cover them to any extent. It is not probable that the crystallines in Connecticut had been brought nearer to baselevel than those in New Jersey at the time of the Cretaceous deposits. There is no evidence to show that the subsidence was greater in Connecticut than in New Jersey, and, therefore, from *a priori* considerations, the conclusion would seem to follow that the subsidence, which permitted the Cretaceous sea to cover the Triassic sandstone area of New Jersey, was not sufficient to permit the sea to cover the then unsubdued crystalline hills of Connecticut. Although this hypothesis is not to be hastily thrown aside, for theoretical reasons, yet it would seem

necessary to hold it very lightly, at least until some positive proof is found of the former existence of the Cretaceous or some later formation in that region. The first suggestion, that the lower Connecticut was a consequent river in the Cretaceous cycle and was revived by the post-Cretaceous uplift, is, at the present state of knowledge, the most probable.

The Farmington. The roundabout course of this river presents another interesting problem, which is not free from difficulties. From its source in Massachusetts it flows southeast across the crystallines to the village from which it takes its name, where it turns abruptly north along the Triassic sandstones for ten or twelve miles, when with another wide sweep it crosses the trap ridges at Tariffville by a deep gorge, and resumes its southeasterly course to the Connecticut. Of this latter part I will speak later, but now arise the questions, “what has been the history of this river,” and “why does it turn north at Farmington?”

The Farmington in the Tertiary cycle. A course more accordant with the structure would seem to be south along the Quinnipiac and Mill river valleys to the sound at New Haven. As has been said before (page 376), Prof. Dana has expressed the opinion that the gorge at Tariffville was occupied by the Farmington in Tertiary times, and that the Westfield river gap further north and the gorge of the Quinnipiac southwest of Meriden are also of earlier date than the glacial epoch. One reason has also been given why I differ from him in regard to the Quinnipiac and Tariffville gorges—they are narrower and steeper than those made in similar rocks during the Tertiary cycle. But more than this, the constructional topography, resulting from the tilting and faulting of the region, could not, it would seem, have caused the Farmington to take its present course. Even if it had taken this roundabout course during the baseleveling of the country, it must, since it would have had to cross three trap sheets, have been captured and led to the sea by the shorter and easier way along the sandstone area. The fact that the Connecticut probably persisted in its consequent course is no argument for similar conditions for the Farmington, because the latter is much the smaller stream, and so more easily captured. Nor could the river have been forced into this course during or after the post-Cretaceous uplift, for the land was then raised more at the north than at the south, and any changes from this cause would have been to confirm the river in its southward course. It is very probable, therefore, that in at least the latter part of the Tertiary cycle,

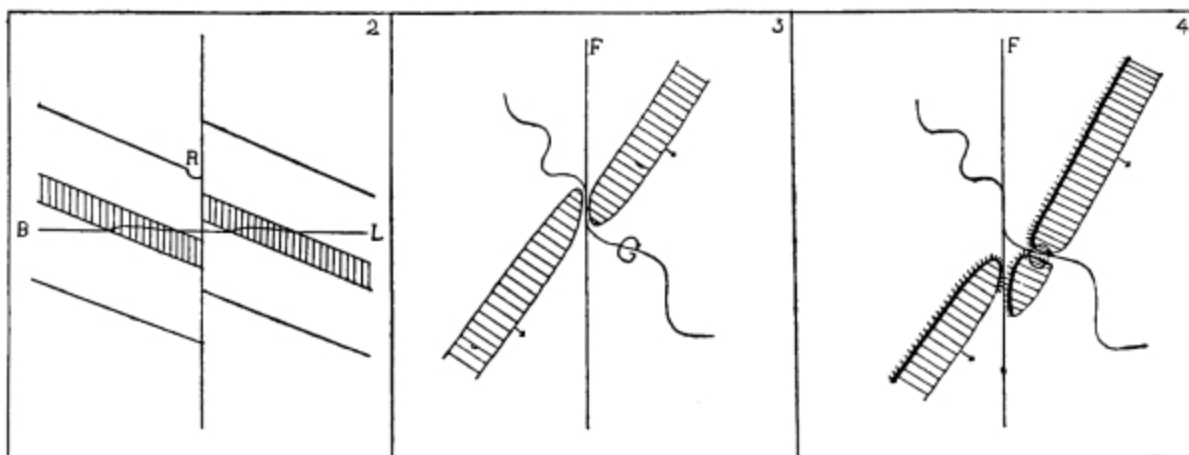
the Farmington did not have its present course, but followed the open sandstone valley, along the course of the Quinnipiac and Mill rivers of to-day. The earlier history of this river is purely conjectural; one fact may shed a little light upon it, a fact which may indicate that this course was an adjusted one taken during Tertiary times.

In pre-Tertiary times. Origin of Cook's Gap. A few miles southeast of where the river emerges from the crystallines, the trap ridge is cut by a deep wind notch—Cook's Gap—through which the New York and New England Railroad passes west from New Britain. As was pointed out some time ago by Prof. Davis,⁴⁷ this is not a fault gap, because the alignment of the ridge is not broken, but it is probably an abandoned water gap, the head-waters of the stream which formerly occupied it having been abstracted by a rival, which did not have to cross a hard trap ridge. Perhaps this river was the ancestor of the present Farmington, and in that case its history would seem to have been as follows. A stream consequent upon the constructional topography after the faulting and tilting at the close of the Triassic, it had its upper course on the crystallines, its lower on the sandstones and buried trap sheets. In its old age it crossed by a shallow gap the trap sheet, which had been uncovered by erosion. In the second or Tertiary cycle it was simply a revived stream quickened to a new life by the post-Cretaceous uplift of the peneplain. This uplift gave opportunity to a rival stream, which did not have to cross the hard trap beds to intercept the waters of the old Farmington, and lead them out by a shorter, easier path, probably down the sandstone valley west of the trap ridge. The path across the trap was abandoned, and the notch became a wind gap; the river following its new course, until the incursion of the ice-sheet interrupted its normal development. This is of course almost entirely speculative. Cook's Gap is best explained as an abandoned river gap; the Farmington is the nearest river of a size proportional to the size of the gap, and the hypothesis is a rational one. There is, however, no direct evidence that the Farmington once occupied Cook's Gap.

The Tariffville cut. Before attempting to answer the second question, "why the river flows north at Farmington?" let us consider for a moment the history of the Tariffville cut. The river occupies a gorge whose sides are steep and talus covered, but which is not at all clogged with drift. There is naturally no room at or near the water level, even for the wagon road, place

for which has been blasted near the top of the gorge. The profile of the gap shows a gentle ascent from the top of the gorge, up to the nearly level crest line of the ridge. That is to say, the recent gorge has been cut in the bottom of a sag in the ridge. We have already given our reasons for believing that the gorge here is much younger than the Westfield river gap; that it is a part of the work of the next cycle; that it is post-Tertiary. The sag, however, in the bottom of which the gorge is cut, is clearly of the earlier cycle. The bottom of the sag is much above the level to which the rivers had cut their valleys in the late Tertiary, and, therefore, it is certain that a river could not have occupied it at the close of that cycle. It was probably an abandoned water-gap whose stream had been captured in the same way and in the same cycle as the river, which formerly occupied Cook's Gap.

The fact that the sag and gorge, although located very near a fault line, do not correspond to it, but are transverse and independent of it, is instructive and needs a moment's attention. It seems probable that the stream consequent upon the faulted blocks would have flowed down the slope of the tilted block and then along the fault line at the foot of the fault cliff and would have held this course during the baseleveling of the country. When the area was baseleveled the stream must have swung from side to side in its broad flood plain, and thus departed from the fault line. When it was revived by the post-Cretaceous uplift, it was confined to the course it had unwittingly taken on the sandstones just above the hard ridge, and it was forced to cut down through the trap. Subsequently a rival, which did not have to work against this obstacle, abstracted its head waters and the gap was abandoned. The accompanying diagrams may make this easier to understand. [Figure 3](#) is a cross-section of the faulted monocline, R showing the position of the river along the foot of the fault cliff. The line B L represents the surface of the country after baseleveling, the trap outcrops forming *low* hills (much exaggerated in the diagram). [Figure 4](#) shows the dislocated trap sheets, the fault line and the winding course of the river, which has abandoned the fault line except where it passes between the low trap hills. Here the country is at baselevel. [Figure 5](#) represents the region after the elevation and resulting erosion. The trap ridges have become more pronounced, and have migrated eastward in the direction of the dip. The river has been slowly let down upon the northern one from the sandstone at point G and has there cut into the solid trap.



FIGURES 3-5.

The transverse notch of Cook's Gap, already described, was probably located in a somewhat similar manner, but the case is not so clear as at Tariffville.

Gravel terraces of the Farmington. A consideration of some facts concerning the height and slope of the terraces along this part of the river may give a clue to the answer to our question. One-half a mile east of Tariffville and east of the trap ridge, the highest terrace is 210 to 215 feet. Half a mile south of the same place but west of the ridge the height is 275 feet.⁴⁸ The top of the gorge at Tariffville is about 190 feet above the sea-level. It does not seem probable that these highest terraces were ever continuous over all the Farmington valley. But if they represent the level reached by the maximum flood accompanying the melting of the glacier, the great difference in their height on the two sides of the trap ridge, in connection with the other evidence already noted, gives strong reason for believing that the gorge as it exists to-day had not then been cut. A mile and a half east of Tariffville there is a lower terrace which is wide-spread. Its general height is about 190 feet, in places a little more. In this terrace the lower part of the Farmington has cut a trench 90 to 100 feet deep. The shape of the valley makes clear the fact that before this trench was cut the river flowed at about the 190 foot level, which is the height of the bottom of the sag at Tariffville. On the west side of the trap ridge there is also a more or less wide-spread terrace at about the same height. It seems very probable

therefore that the river was raised to the level of the old sag in the trap ridge by the building of these terraces.

The present average southward slope of the highest terraces west of the trap ridge from Northampton, Mass., to Farmington, Conn., forty-four miles, is seven inches per mile,⁴⁹ and Professor Dana is inclined to believe that this is approximately the slope at the time the terraces were built. The character of the deposits shows that the current which formed these deposits flowed south. The present river, flowing north, falls twenty feet between Farmington and Tariffville, or $1\frac{2}{3}$ feet per mile. The reversal of the river was probably determined by two factors. Near the village of Farmington, the waters of 200 square miles of territory are poured into the valley by the upper Farmington and its tributary, the Pequabuck. During the terrace building stage the great mass of *débris* contributed by these streams was deposited where the steep gradient of the highlands was exchanged for the gentle slope of the lowland. The main north and south valley was thus choked by the *débris* of its tributaries and a long stretch of comparatively still water extended north from Farmington, in which nearly horizontal deposits were made. South of Farmington the terrace deposits are much coarser than to the north, and the face of the terraces is much greater. It is not impossible that, as the deposits between Farmington and the Massachusetts state line approached nearer and nearer to horizontality, the waters of the upper Farmington began to divide, part flowing north and part south, the northward flowing portion finding an outlet at the sag at Tariffville. If this was the case, the terraces between Farmington and Tariffville must have had a slight slope to the north. Their present southward slope could readily be accounted for by the re-elevation of the land after the disappearance of the ice. This explanation rests upon the ability of the upper Farmington and the Pequabuck to have completely dammed the southward flowing current and turned it northward by the great mass of their deposits. If this was not the case, and there may be some doubt on the matter, the subsidence which accompanied the later stages of the ice-retreat is the other factor in the problem. It is estimated that an average depression of 1.25 feet⁵⁰ through the Connecticut valley would restore it to an altitude approximating that at the close of glaciation. It seems highly probable that these terrace-deposits were built before the maximum depression was reached. If this was the case, the depression

would be efficient in reversing the Farmington, and this factor would supplement the first. It is impossible at present to say to what extent these two factors enter into the problem. That they are not mutually exclusive is evident, and that they are together quantitatively competent seems certain. Among the several hypotheses which have been considered, this seems the most probable, and in the light of the present evidence the most rational.

At first thought it might seem that if the Farmington was reversed by the differential subsidence of the land, the Connecticut ought to have suffered a similar fate, and since it did not, the explanation cannot apply to the Farmington. But the terraces of the Connecticut have a much greater southward slope than those on the smaller river, and the depression was not sufficient to reverse the stream. The conditions on the two sides of the trap ridge were not the same.

To sum up, then, the history of the Farmington seems to have been as follows: Its original consequent course was southeast on the crystallines and perhaps across the trap ridge at Cook's Gap, from which course it was turned in the Tertiary cycle by a stream whose course was approximately that of the Mill river of to-day. The damming of the valley by the deposits of the Upper Farmington, and the depression in the north accompanying the ice retreat, reversed the river at Farmington, and it took a new course on the terrace deposits, escaping by the sag in the trap at Tariffville into the Connecticut valley.

The Quinnipiac. The gorge of the Quinnipiac, already mentioned several times, seems closely comparable to the gorge of the Farmington. It is not of the Tertiary cycle, and is best referred to the inter-glacial or post-glacial epochs. We should expect the Quinnipiac, instead of turning eastward, to cut through this sandstone ridge, to continue southward along the Mill river valley. Dana⁵¹ finds from the heights of the terraces that the drainage of the terrace-building period was not along the Quinnipiac, but along the Mill river, and concludes that the Quinnipiac gorge was obstructed by an ice-dam. I have not as yet studied it enough in detail to do more than express the opinion here reiterated, that this gorge is later than the cycle in which the open sandstone lowland on either side of it was excavated. Its topographic form would put it in the cycle which has been called post-Tertiary.

The Scantic. In the Scantic we have a typical example of a river whose lower course is manifestly of a later date than the upper. In this it is similar to several of our Atlantic rivers, notably those of North Carolina, whose upper courses are on the Piedmont crystallines, being probably established previous to the Cretaceous baseleveling, and whose lower courses stretch seaward over the unconsolidated Tertiary deposits of the coastal plain. As the plain of these recent deposits emerged from the sea, the rivers were forced to extend their courses eastward over the freshly raised surface to the retreating shore line. The Scantic river has a similar history. Its upper course in southern Massachusetts on the crystalline plateau is a remnant of the drainage established before Cretaceous baseleveling and revived by the subsequent uplift. How much that revived drainage has been modified by drift can only be determined by long field study, but the topography, as read from the topographical atlas would seem to indicate, that it has not been much. The valleys were undoubtedly clogged with drift, and the drainage area may be somewhat modified, but the drainage seems to be substantially along the same lines.

Just below the village of Hampden, the Scantic leaves the plateau and enters the Triassic lowland. From this point to its mouth at the Connecticut, opposite Windsor, a distance of twenty miles, it flows nearly all the way through the gravel, sand and clay deposits of the period of ice-retreat. The topography of the lower course of the river is entirely characteristic of a stream which has recently attacked a level, easily eroded district. The inter-stream surfaces are broad and flat; the descent to the stream bed which is sunk seventy or eighty feet below the general surface is exceedingly steep. These two lines, that of the inter-stream surface and that of the valley side, meet at a sharp angle. The side streams are as yet very short, and have cut narrow gorges down to the main river. Tributary to them are deep side ravines, whose bottoms ascend rapidly to the inter-stream surface, the whole making a dendritic system of drainage in its earlier stages. The Scantic, having reached base level in its lower course, has developed a narrow flood-plain.

Manifestly this part of the river valley is of much later date than the upper part. If, during the period of ice-retreat, the lower Connecticut valley was an estuary, the Scantic was a much shorter river than at present. Its mouth could not have been far from the point where now it leave the

crystallines, but as the land was elevated and estuarine conditions gave place to fluvial, the Scantic lengthened "mouthward," consequent upon the minor inequalities of the newly made beds. The effect would be substantially the same if the terraces were built by great valley floods, as Dana supposes. In pre-glacial times this river, in common with several other rivers rising on the crystallines and flowing into the Connecticut, had courses of various lengths over the Triassic sandstones, but these old valleys are lost entirely, the later trenches in the terrace deposits being altogether independent of them.

Other examples. The lower Hockanum, Farmington, Park, and the entire length of many short streams are similar to the lower Scantic, and originated under similar conditions. Stony Brook, a little stream north of Windsor Locks, presents the same features, but with this variation: It is superimposed through a thin layer of drift upon the sandstone, into which it has cut a deep, picturesque gorge. The Hockanum and Farmington are also "locally superimposed" in a few places. The Connecticut, also, north of Middletown, although following its pre-glacial valley, has departed in numerous places from its former bed, and has cut down through the valley-filling onto ledges of rock beneath. The water-power at Enfield, Conn., and at Turner's Falls and Bellows Falls, Mass., is the result of this superimposed position.

Abandoned gaps. Many abandoned water-gaps must exist among the hills of the state. Cook's Gap, through which the New York and New England Railroad crosses the trap ridge, three miles west of New Britain, has already been discussed. It must not be confounded with the majority of the other gaps in the trap ridge, which are oblique, break the alignment of the ridge, and are due to faults.

The New York and New England Railroad in ascending to the eastern plateau passes through Bolton Notch, a few miles east of Manchester. This notch, also, is an abandoned river bed but, as it seems, abandoned at a later date and for another reason than that assigned for Cook's Gap. The drift is very heavy in this region, and the most probable explanation is that the post-glacial streams do not altogether follow pre-glacial valleys. This gap, used by turnpike and railroad, testifies of another and older drainage system.

That in this brief article all the problems connected with the Connecticut rivers have been solved, or even noted, is not to be expected. It is hoped, however, that the work done may prove a help to further study of the same regions, and that the tentative conclusions advanced may be substantiated by further investigation.

HENRY B. KUMMEL.

FOOTNOTES

³³ The author desires to express his obligation to Professor W. M. Davis for aid in the preparation of this article. It was first written under his direction and with the help of his suggestions when the author was in the graduate school of Harvard University. Prof. Davis is not responsible, however, for the statement of the views herein advanced, although in general it is believed that he is in accord with them.

³⁴ Amer. Jour. Sci. 3d ser., vol. xxxvii, 1889, p. 423. Bull. Geol. Soc. Amer., vol. ii, p. 545.

³⁵ The rough diagrams accompanying this paper may aid the reader who is unacquainted with the details of the region under discussion. The abbreviations on the [above figure](#) are as follows: C. The Connecticut. Cr. Pl. Crystalline plateau (the shaded area). F. The Farmington. H. Hartford. Ho. The Housatonic. Lm. Limestone area. M. Meriden. Mi. Mill River. Mt. Middletown. N. The Naugatuck. N. H. New Haven. No. The Norwalk. Q. The Quinnipiac. Qg. The Quinnebaug. S. The Scantic. Sa. The Saugatuck. T. Tariffville. Th. The Thames. The unshaded area is the Triassic sandstone lowland, and the blackened areas represent the ridges of the faulted trap sheets.

³⁶ Am. Jour. of Sci., 3d ser., vol. xxxvii, p. 430.

- ³⁷ Bulletin of Geol. Soc. of Amer., vol. ii, p. 554.
- ³⁸ It is not desired to affirm that these periods of erosion and elevation began and ended promptly with the beginning or end of a period. The time statements must be considered as only approximate.
- ³⁹ An exaggerated idea must not be had of the steepness and narrowness of these crystalline valleys. The valley of the Farmington, five miles up from where it opens into the Triassic sandstone, is 400 to 500 feet deep, and a mile and a half wide at the top. The Connecticut valley, just below Middletown, is about 400 feet deep and two miles wide at the top. These are fair representatives of the valleys in the crystalline rocks in the central part of the state.
- ⁴⁰ Amer. Jour. of Sci., vol. x, 3d ser., 1875, p. 506.
- ⁴¹ McGee. Amer. Jour. Sci., 3d ser., vol. xxxv, p. 376.
- ⁴² Davis and Wood, Geographic Development of Northern New Jersey, pp. 413, 414.
- ⁴³ "Rivers and Valleys of Pennsylvania," Davis, W. M., published in The National Geographic Magazine, in 1889.
- ⁴⁴ Davis, W. M., "Geographic Development of Northern New Jersey," p. 397-8.
- ⁴⁵ Davis, W. M., Amer. Jour. of Sci., 3d ser. vol. xxxvii., 1889, p. 432.
- ⁴⁶ Geog. Devel. of Northern New Jersey, p. 404 et seq. Proc. Bos. Soc. Nat. Hist. Also Rivers of Northern New Jersey, p. 11 et seq. National Geographic Magazine, vol. ii, p. 93.
- ⁴⁷ Faults in the Triassic Formation near Meriden, Conn. Bulletin of the Mus. Comp. Zoöl. Harvard Univ. vol. xvi. No. 4, p. 82.
- ⁴⁸ J. D. Dana, Amer. Jour. Sci. 3d. ser., vol. xv, p. 506.

[⁴⁹](#) J. D. Dana, Amer. Jour. of Sci. 3d ser. vol. xxv, p 446.

[⁵⁰](#) J. D. Dana, Amer. Jour. of Sci., 3d ser., vol. xxiii, p. 198.

[⁵¹](#) J. D. Dana. Amer. Jour. Sci., 3d ser., vol. xxv, p. 441.

STUDIES FOR STUDENTS.

GEOLOGICAL HISTORY OF THE LAURENTIAN BASIN.

The study of the Pleistocene history of the basin drained by the St. Lawrence has been fragmentary and is still far from being complete. There is a lack of agreement in the interpretation of observations already made, due in part to the comparatively limited portion of the field examined even by those who have given the subject most attention, and in part to lack of uniformity in the standards of comparison used. It is with the hope of assisting in reaching more harmonious results that attention is here invited to methods of study.

In the present treatment of the subject it may be advantageously subdivided, and the facts and hypotheses relating to each division separately considered. Of the divisions that may be suggested the following seem the most important:

1. Character of the sub-morainal or hard-rock topography in the Laurentian basin.
2. Origin of the basin.
3. Sedimentary deposits.
4. Shore markings left by former water-bodies.
5. Fossils in ancient sediments, shore ridges, terraces, etc.
6. Fauna of the present lakes.
7. Changes in elevations of the land.
8. Former outlets.

9. Probable effects of an ice sheet on drainage.

10. Probable effects of a subsidence which would make the basin an arm of the sea.

1. *Character of the hard-rock topography.* In order to learn the character of the Laurentian basin it is necessary to examine the rock surface beneath the general covering of glacial débris and stratified sediments which partially fill it. To do this, those areas in which rock in place forms the surface require to be mapped and their elevations noted; the records of wells and other excavations which pass through the superficial deposits should also be obtained and the character of the underlying rock ascertained, as far as is practicable. When sufficient data of this nature shall have been recorded, a contour map of the basin can be drawn that will reveal the shape of the depression with which the student has to deal. The depth of the present lakes plus an estimated thickness of clay and morainal material covering their bottoms, will probably furnish the only means of sketching contours over the deeper portions of the basin. Even an approximately accurate map of this character cannot be constructed for a long time to come, but every advance towards it will serve to make the problems to be studied more and more definite.

Something of the form of the rock-basin is already known and several deep channels in its borders, now filled with drift, have been discovered. The courses of buried channels connecting the basins of some of the present lakes have also been approximately determined. It is not necessary at this time to refer specifically to the discoveries that have been made, but it may be stated that enough is known to assure us that the basin is a depression in solid rock, the bottom of which is below sea level.

2. *Origin of the basin.* The rocks in which the Laurentian basin is situated are, with the exception of the Lake Superior region, nearly horizontal and belong almost wholly to the Paleozoic. The basin is essentially a depression in undisturbed strata, and all who have considered its origin seem agreed that it has been formed by excavation. A vast mass of horizontal strata has been removed, leaving an irregular rim of undisturbed rocks on all sides. The form of the depression is now obscured by drift; the deeper portions contain stratified sediments which have been deposited within it and it has been warped somewhat by orographic movement.

The manner in which the excavation was formed has been explained principally in two ways. One hypothesis is that it owes its origin to a time of subaërial denudation preceding the Glacial epoch, during which a valley, or series of valleys, was worn out by stream erosion; and that the depression thus produced has been but slightly modified by ice action. The closing of the ancient valley has been referred to orographic movements and to the filling of its outlet by glacial débris. Another hypothesis is to the effect that the excavation is mainly due to ice erosion during the Glacial epoch, without special reference to previous topographic relief. A warping of the earth's crust so as to produce a true orographic basin does not seem to require consideration, for the same reason as already stated, that the rocks in which the basin lies have been but little disturbed from their original horizontal position. Future study of the region must determine which of the two hypotheses outlined above best suits the facts; or if each hypothesis has something in its favor, what combination of the two may be accepted as the final explanation.

It is a suggestive fact in connection with the first of these hypotheses, that the youngest rocks in the region antedating the Pleistocene belong to the Carboniferous. This seems to show that the land has not been submerged since at least the close of the Paleozoic. If not a region of sedimentation during this vast interval, it must have been subjected to erosion. The erosion of an ancient land surface might result in the production of topographic forms of diverse character, depending on its altitude, on the length of time it was exposed to atmospheric agencies during various stages of elevation, and on climatic and other conditions. The study of topographic forms is now sufficiently advanced to enable one to predict somewhat definitely what features would appear under certain conditions. We also know the characteristics of topographic forms due to glacial erosion. It seems evident, therefore, that a knowledge of the hard-rock topography in the Laurentian basin, would enable one to draw definite conclusions in reference to the part that ice and water each had in shaping the forms now found there.

The conclusion that the region under consideration has been glaciated is well established; it remains, therefore, to determine what topographic forms, if any, due to pre-glacial stream erosion can be recognized. As an example of this kind of evidence desired, attention may be directed to the

northward facing rock escarpments which follow the southern shores of lakes Erie and Ontario for a large part of their courses and at varying distances up to several miles. These escarpments are composed of the edges of nearly horizontal strata, mostly of Paleozoic limestone, and their bases are buried beneath glacial débris and stratified clays so deeply that in some instances, at least, they do not reveal half of their actual height. These escarpments not only have Pleistocene deposits banked against them, but their faces and summits are polished and grooved, showing how stubbornly they resisted the invasion of the ice which impinged against them from the north. South of lake Ontario especially, the trend of the escarpment referred to is directly athwart the course of the ancient glaciers. The entire history of these escarpments cannot be discussed here, as my desire is simply to call attention to the fact that they existed before the Glacial epoch, and are relics of a strongly accented pre-glacial topography. They are within the southern border of the Laurentian basin, and hence afford means of determining, in part, what was the form of that basin before it was modified by ice action. Other similar escarpments exist in the northern and western portions of the same great basin, and as this study progresses it is to be expected that still other features of the pre-glacial land will be revealed. It is perhaps too early to decide what were the special topographic forms which gave character and expression to the St. Lawrence basin before the ice invasion, but the Erie and Ontario escarpments and some other similar features now recognized, suggest that in Tertiary times it resembled the present condition of the upper portion of the Mississippi valley, where bold, rock escarpments border wide stream-worn depressions.

Deep drift-filled channels are known to cut across the Erie and Ontario escarpments. These seem to have been formed by streams tributary to the main drainage line to the north. If this conclusion is well founded, a study of the hard-rock topography should reveal other similar channels and finally indicate a well matured drainage system. If even the broader and stronger features of the pre-glacial surface can be determined, then the modifications due to glacial abrasion will become conspicuous, and the amount that glaciers have broadened and deepened the basin be determinable.

A study of the lithological character of the drift south of the present lakes should show, at least in a rough way, what portion of it was derived from the waste of rocks within the Laurentian basin. This inquiry has

already been undertaken by at least two geologists, and estimates of the quantity of material removed from the basins of lakes Michigan and Erie respectively, have been made. This method may be extended so as to embrace a larger area, or some special portion of the great depression best suited for the trial may be selected. If the material removed from the basin or re-distributed within it by glacial action can be shown to be approximately equivalent in volume to the amount of rock excavated in order to form the depression, it would evidently tend to support the hypothesis of glacial erosion. If, on the contrary, the amount of débris derived from the basin should fall far short of what would be requisite to refill it, no very definite conclusion would seem to be indicated unless account could also be taken of the fine material carried away by glacial streams.

As the case stands at present it appears that there is evidence of a pre-glacial valley or series of valleys as has been claimed by several geologists, and that all but the boldest features of the old topography have been obliterated or greatly modified by glacial erosion followed by glacial and other sedimentation. Additional observations should show somewhat definitely the amount of work assignable to particular portions of the history. How far the results of subaërial and of glacial erosion have been modified by other agencies, more especially by orographic movements, has also to be considered.

If the St. Lawrence basin shall be shown to be largely the result of subaërial erosion it will follow, unless it is found that the deeper portions are the result of glacial action, that the land at the time the streams did their work, must have stood higher than at present, for the reason that the bottom of the depression is now below sea level. Some idea of the smallest amount of elevation necessitated by this hypothesis might be obtained by estimating the gradients of the ancient streams and the amount of elevation required to bring the bottom of the depression up to sea level.

A study of the hard-rock topography in the valleys of the Ottawa and St. Lawrence and of the present submerged Atlantic border of the continent would also be instructive in this connection. The strict correlation of the topographic history of the interior and of the continent's margin may be difficult, but as the two regions are directly connected, valuable results should follow their comparative study.

The hypothesis that the Laurentian basin is due largely to pre-glacial erosion, necessitates also that the ancient system of river valleys should have been closed in some way so as to form the basins of the present and of former lakes. The closing has been referred to several agencies. An unequal subsidence following the period of stream erosion has been postulated. During the Glacial epoch the entire region was ice-covered and only glacial streams of one kind or another could have existed. On the retreat of the ice, when portions of the basin were abandoned, the drainage is supposed to have been obstructed by the ice itself, as will be noticed below. When the glaciers melted, a vast sheet of débris was left which in many instances filled or obstructed previous drainage lines. Old channels, now deeply buried, have been reported to connect the basins of the various existing lakes, as has already been mentioned, but no similar channel which could have afforded an escape for the waters of the entire basin has been discovered. Here again an acquaintance with the hard-rock topography should give assistance and indicate either that such a channel existed or that orographic movements have taken place which have obstructed the former drainage system. The glacial hypothesis assumes that the basin was excavated mainly by glacial abrasion and does not require that the land should be either higher or lower than at present. The study in this direction merges with that of the general glaciation of the northeastern part of the continent, and cannot be treated at this time.

3. *Sediments*.—Regularly stratified deposits of clay and sand occur along many portions of the borders of the present Laurentian lakes. These were clearly formed in water bodies which formerly existed within the Laurentian basin, and which in certain directions, at least, were of wider extent than the present lakes. The areas occupied by these deposits have been partially mapped, but much remains to be done in this direction. Fresh sections, particularly of the stratified clays, are exposed from time to time by artificial excavations, in which much of their history may be learned. Not only should records be made of the facts noted at special excavations, but the extent and character of the stratified deposits in one area should be determined and compared with similar data obtained in other areas. For example: the clays covering large tracts on the west shore of Lake Michigan and on the southern and western border of Lake Superior are of a red color, while other areas bordering Lake Erie are covered with blue clay. These two deposits have been supposed to have been laid down at the same time and

in the same lake. The definite correlation of the clays of these two areas by direct contact, however, does not seem to have been made, and there are reasons for thinking that they may be quite distinct and that they originated in separate lakes.

The outer limits of the deposits of clay and sand here referred to are known in some instances to be determined by ancient beaches and terraces. Such associations of deep and of shallow water deposits require special attention, as the study of one may assist in interpreting the significance of the other. The fine, evenly stratified clays frequently contain large angular boulders, which appear to have been dropped from floating ice and to show an intimate connection between the ancient lakes and neighboring glaciers. The possibility, however, of the boulders having been brought into the ancient water bodies by rivers, or floated outwards from the shore by lake ice, should also be considered. Huge angular masses of limestone have been reported as occurring in southern Michigan especially, which rest on superficial deposits and are thought to have been carried northward by lake ice. The relations of these masses to well defined shore lines have never been determined. If it should be found that they are above all former shores, it is evident that they must have been carried by some other agency than the one mentioned.

A chemical examination of the clays, or of their contained water, may indicate whether or not the basin was formerly in direct communication with the ocean. Analyses of the clays of the Champlain valley and of the similar clays in the Ontario and Erie basins might indicate whether or not they were deposited under similar conditions.

4. *Shore records.* Beaches and terraces have been studied at many localities about the borders of the present lakes, sometimes at a distance of more than twenty miles from their margins and at various elevations up to several hundred feet above their surfaces. In some instances these ancient shore records have been followed continuously for scores of miles. The tracing and mapping of individual beaches is one of the most important parts of the study here outlined, and is already well advanced. Confusion has unfortunately arisen, however, for the reason that topographic features, due to shore action, have, in some instances, been confounded with somewhat similar features due to other causes. Moraines and gravel ridges,

formed by glacial streams, have been mistaken for beach ridges, and terraces of various origin have not been clearly discriminated.

In order not to be led astray by topographic forms that simulate shore phenomena, the student should examine the shores of existing lakes and learn what records are there being made. In the study of topography, “the present is the key to the past,” just as definitely as in any other branch of geology. The topography of lake shores has already received attention from one skilled in reading geological history in the relief of the land⁵² and the study of existing shores in the light of what has already been done in that direction should enable even the beginner to avoid falling into serious error in interpreting ancient records of the same nature.

To be able to discriminate clearly between shore features and somewhat similar glacial phenomena, it is necessary to become familiar also with the topography of glacial deposits. Fortunately in this study also a guide is at hand⁵³ which, in connection with field observations, should soon train the eye to discriminate the shapes assumed by moraines and the deposits of glacial streams from all other topographic forms.

In examining the records of former lakes it will soon be observed that, in many instances, where the highest of a series of ancient beaches is obscure and indefinite, the topographic expression above and below a certain horizon, and also the character of the surface material, whether of the nature of lacustral clays and sands or of glacial débris, residual clay, etc., above and below the same level, are significant, and enable one to map the outline of a former water body with considerable accuracy.

In tracing ancient beaches and terraces, their forms and internal structure need to be recorded, so that the fact of their being true shore records may be made plain to others. The elevations of various well-defined points throughout the extent of an ancient shore should be carefully measured, for, as will be noticed below, although originally horizontal, they have, in many instances, been elevated or depressed, owing to broad general movements of the earth's crust. The continuous tracing of individual shore lines for as great a distance as possible is highly desirable, especially in a wooded country, in order to be positive as to which ridge or terrace measurements of elevation relate, and also for the purpose of observing the nature of the changes that occur when a shore line gives place

to other records. For example: some of the ancient beach ridges about the west end of Lake Erie have been found to be continuations of moraines. In other instances shore ridges have been reported to end indefinitely and to be replaced at the same general horizon by glacial records of various character. The correct interpretation of phenomena of this nature is especially important.

Accurate measurements of the vertical intervals between well defined beaches at many localities would enable one to identify special horizons, providing orographic movements were not in progress during the time the series was forming. This method has recently been successfully applied on the north shore of Lake Superior, where the character of the country does not admit of the tracing of individual terraces for considerable distances.

The deltas of tributary streams should also be revealed in the topography of the basin of an ancient water body. Changes in the character of lacustral sediments near where rivers emptied are also to be looked for. Sand dunes are frequently an important accompaniment of existing shores, and their association, perhaps, in a modified form, with ancient beaches is to be expected.

5. *Fossils.* Thus far only a few fossils have been found in the stratified clays and sands or in the ancient beaches of the Laurentian basin. Such observations as have been made in this connection indicate an absence of the remains of marine life and the presence, in a few instances, of fresh-water shells in all of the basin west of the eastern border of the basin of Lake Ontario. To the eastward of Lake Ontario, however, in the St. Lawrence and Champlain valleys, marine fossils are common in deposits supposed to be contemporaneous with the stratified clays to the west.

A careful search in the clays and beaches left by the former water bodies might be rewarded by important discoveries. In this examination microscopical organisms should not be neglected. If after a detailed examination no fossils are discovered, this negative evidence would have its value, as it would indicate that the physical conditions were not favorable to life, and an explanation for this fact might be found. It is scarcely necessary to mention that care should be taken not to mistake the shells occurring in modern swamp deposits associated with the ancient beaches for true lacustral fossils.

About the borders of the present lakes and sometimes even below the level of the lowest of the ancient beaches the remains of the mastodon, elephant, giant beaver, elk, bison, deer, etc., have been found. The recency of the existence of such of these animals as are extinct may thus be established, as well as the former distribution of those still living in other regions.

Evidence of the existence of man has been reported from one of the old lake ridges in New York, and it is important that this interesting discovery should be sustained by evidence from other localities. Stone implements especially should be looked for in undisturbed lacustral clays, and in the gravels of the ancient beaches.

The remains of forests have been stated to occur in the lacustral clays adjacent to the south shore of Lake Erie. It is desirable to know the extent of these deposits and how continuous they are; also the character of the plant remains they contain, and whether they have been disturbed from the position in which they grew. Some of the questions that may be asked in this connection are: Was the basin drained and forest covered before the vegetable remains were buried, or were the plants floated to their present position, or did they grow on moraines covering the stagnant border of the retreating glacier and become involved and buried in morainal material as the ice melted?

6. *Life in the present lakes.* The fauna of the present lakes has a bearing on their past history, for the reason that in the deeper parts of lakes Superior and Michigan crustaceans and fishes have been found which are believed to be identical with marine forms. These may be considered as “living fossils,” and are thought by some to indicate that the lakes in which they occur were formerly in direct communication with the ocean. If the occurrence of living marine species in the present lakes is found to be widely at variance with the history of the basin as determined from physical evidence, an inquiry should be made in reference to the manner in which the species discovered might migrate.

7. *Changes in elevation.* One of the most difficult problems in connection with the history of an inland region is the determination of changes of level. By leveling along an ancient beach, post-lacustral changes in the relative elevations of various points may be readily ascertained. Pre-

lacustral changes, however, by which ancient valleys have been obstructed, are much more difficult of direct observation, but might appear from the study of the hard-rock topography, as has already been suggested. This branch of the investigation, however, should more properly begin at the coast and be extended inland.

8. *Former outlets.* Several localities where the waters of the Laurentian basin have overflowed during former high-water stages have been pointed out, but some confusion has arisen in this connection, for the reason that the channels formed by streams issuing from the margin of the ice during the closing stages of the Glacial epoch have, in some instances, been mistaken for evidence of former lake outlets. The old outlets which seem to have been well determined are situated at different levels, and show that the entire basin could not have been occupied by a single great water-body, unless, as has been supposed by some, it was in direct communication with the sea. This hypothesis will be considered below. It has sometimes been assumed that all of the basin below the level of some ancient outlet was once flooded, so as to form a great lake in all of the basin now situated at a lower level; but, in making such generalizations, the possibility of places in the rim of the basin being at a lower level than the outlet discovered, thus necessitating a special explanation, such as the partial occupation of the basin by glacial ice, or changes in elevation of such a character as to raise the locality of former overflow or to depress other regions, have to be considered.

Former outlets should bear a definite relation to neighboring shore lines and to sedimentary deposits. The channels leading from former points of discharge merit examination, as here again changes of level may perhaps be detected in the gradients of stream terraces.

Most of the ancient outlets thus far recognized lead southward, but as previously mentioned, a former channel of discharge north of Lake Superior has recently been reported. If this observation is confirmed, it will have an important bearing on questions relating to changes of level and to the position of the ice front during the later stages in the retreat of the glaciers.

9. *Probable effects of a retreating ice sheet on drainage.* The generally accepted conclusion that glaciers advanced southward and occupied the Laurentian basin during the Glacial epoch and retreated northward toward

the close of that epoch, is sustained by a vast body of evidence. As the ice sheet withdrew it left a superficial deposit frequently one or two hundred feet thick over nearly all of the region it abandoned, and pre-glacial drainage lines were obstructed and mostly obliterated. As long as the slope in front of the ice was southward, the drainage from it found ready means of escape, but when the slope was northward towards the ice front, the drainage was obstructed and lakes were formed.

We have good reasons for believing that the topography of the Laurentian region was essentially the same at the close of the Glacial epoch as it is now, but the broader question of continental elevation is less definite. The inequalities of the surface being essentially as we now find them, it would follow that the first lake formed when the ice retreated to the north of the divide running through central Ohio and central New York, would be small and dependent on minor features in the relief of the land, and would discharge southward. As the ice retreated, the lakes would expand and become united one with another and the larger lakes thus formed would still find outlet across the southern rim of the basin. As the glaciers continued to retreat lower and lower, passes would become free of ice and the lakes would be drained at lower levels, old beaches would be abandoned, the lakes would contract, and finally separate lakes would be formed in the lowest depression in the basins of the more ancient water bodies. The shape of the retreating ice front would be determined by topographic conditions and would in turn determine the northern outline of the lakes along its margin. This in brief is one hypothesis that has been proposed to explain the varied history recorded by the shore records, sediments, etc., within the basin.

10. *Communication with the sea.* Another hypothesis which assumes to account for some of the facts observed, is that the continent was depressed at the close of the Glacial epoch sufficiently to allow the sea to have access to the Laurentian basin. This hypothesis is coupled with others which do not recognize a period of Pleistocene glaciation, but, as already suggested, this is a matter that is considered by the great body of American geologists as not being any longer open to profitable discussion.

In the study here outlined the question whether the water bodies which formerly occupied the Laurentian basin were lakes or arms of the sea, should not be difficult of direct and positive determination. If fossils can be

found within the basin, they might yield definite testimony, but even if they are absent or if their evidence is inconclusive, topography can be appealed to with the expectation of receiving a conclusive decision.

If the Laurentian basin was occupied by an arm of the sea during various stages in the Pleistocene elevation, then the records of such a submergence should occur both within and without the depression, and direct connection between the two should be expected. If the waters within the basin were capable of making such well-defined shore records as are now found, we are justified in assuming that the true ocean beach on the outer slopes of the basin would be still more conspicuous. Again, the waters within the basin deposited a sheet of sediment, certainly not less than one hundred feet thick; to be sure the conditions for rapid accumulation were there present, but if the ocean covered the adjacent land it should have left similar deposits. This is abundantly proven in the St. Lawrence and Champlain valleys, where clays containing marine fossils occur up to a certain horizon and record a Pleistocene invasion of these depressions by the sea. If the adjacent Ontario basin was occupied by the sea about the same time that the Champlain valley received its filling of clays containing marine fossils, there is every reason to believe that the deposits and their contained fossils in each basin would have been essentially the same.

One of the best known of the ancient shore lines about Lake Ontario has an average elevation of approximately 500 feet above the sea. If the sea had access to the basin at the time this breach was formed, then at corresponding horizon without the basin especially, to the south and southeast, where the full force of the Atlantic's waves would have been felt, there should be still more prominent beaches.

Many well-defined shore lines in the Laurentian basin are much higher than the one just referred to, and if these were also formed during various stages of submergence, as has been claimed, it is evident that ocean beaches and ocean sediments of Pleistocene age should be looked for over nearly the whole of the eastern part of the United States. The student may easily answer this question for himself, and thus perhaps make a contribution to the subject here treated.

In the investigation here outlined, the work of previous observers should not be ignored, and every plausible hypothesis that has been

advanced to account for the facts observed should be carefully tested. In writing these pages I have not quoted the writing of others, for the reason that a discussion of evidence has not been the aim in view, and also because the writings examined are so numerous that justice could not be done them in the space at command. That the literature relating to the subject is voluminous is indicated by the fact that an annotated bibliography of the Pleistocene history of the Laurentian basin, now in preparation, already contains over 200 entries of individual papers.

ISRAEL C. RUSSELL.

FOOTNOTES

- [⁵²](#) The Topographic Features of Lake Shores, by G. K. Gilbert, in Fifth Ann. Rep. U. S. Geological Survey 1883–4.
- [⁵³](#) Preliminary Paper on the Terminal Moraine of the Second Glacial Epoch, by T. C. Chamberlin, in Third Ann. Rep. U. S. Geological Survey, 1881–2.

EDITORIALS.

The Summer meeting of the Geological Society of America will be held at Madison, Wis., on August 15 and 16. The session of the American Association for the Advancement of Science will begin at the same place on the 17th of August and extend to the 23d. The Congress of Geologists, under the auspices of the Columbian Exposition, will begin at Chicago, on August 24, and continue its sessions so long as its work may require. Preliminary to this series of meetings, Professors M. E. Wadsworth and C. R. Van Hise will meet such geologists as care to visit the Lake Superior region at the Commercial Hotel, Iron Mountain, Mich., on the forenoon of August 7, and will act as guides during the week following. A carefully prepared scheme for the trip is announced, embracing visits to the leading points of interest in the Menominee, Marquette and Gogebic iron districts, and in the copper-bearing region of Keweenaw Point. Those who desire to participate in the excursion, or who wish information regarding it, should address Professor Van Hise, at Madison.

In connection with the meetings of the Geological Society and the American Association at Madison, there will be excursions to the Devil's Lake region, to the Dells of the Wisconsin, and to the driftless area, under the guidance of geologists personally familiar with the features of most special interest. The article of Professor Van Hise in this number is a timely presentation of some points of peculiar significance in the first named region, and will prove very serviceable to those who choose the excursion to that region.

It is proposed to hold the sessions of the Congress at Chicago at the Art Institute during the forenoons, leaving the afternoons free for visiting the Exposition. Experience has shown that a half day devoted to looking at exhibits, where there is such a plethora of objects of interest as in the Exposition, taxes the faculties of observation to the full extent of their pleasurable employment. Attendance upon the Congress and the study of

the Exposition will, therefore, it is thought, constitute agreeable and profitable complements of each other. Excursions to points of geological interest in the vicinity of Chicago will be privately arranged, if desired.

These three meetings, with the attending excursions and the study of the Exposition, constitute a rare combination of opportunities which will doubtless be embraced very generally by the geologists of the country.

T. C. C.

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The supply of numbers one and two of this JOURNAL remaining in the hands of the publishers has become reduced below the limit they desire to preserve for binding and for special purposes, and they would esteem it a great favor on the part of those who may have received duplicates, as sample copies or by the accidents of mailing while the lists were imperfect, if they would return such duplicates to them. They will gladly return the postage if the address of the sender is placed on the wrapper.

REVIEWS.

CRYSTALLINE ROCKS FROM THE ANDES.

***Untersuchungen an altkrystallinen Schiefergesteinen aus dem Gebiete der argentinischen Republik* von B. KÜHN. Neues Jahrbuch für Min., etc., Beit. Bd. VII., 1891, p. 295.**

***Untersuchung argentinischer Pegmatite, etc.*, von P. SABERSKY, *ib.* p. 359.**

***Untersuchungen an argentinischen Graniten, etc.*, von J. ROMBERG, *ib.*, VIII., 1892, p. 275.**

Travelers and foreign residents in South America are rapidly furnishing information relative, not only to the volcanic, but also to the older crystalline rocks composing the great Andes chain. Since the early observations of Darwin,⁵⁴ the petrographical collections made by Stelzner during his three years' residence, as professor, at Cordova (1873–1876) have been described by himself⁵⁵ and Franke,⁵⁶ while the results of detailed studies of the more extensive collections gathered by Stelzner's successor, Professor L. Brackebusch, are now beginning to appear. Professor Brackebusch's residence in the Argentine Republic lasted from 1876 till 1883, and during this period he made numerous scientific expeditions.⁵⁷ The petrographical material thus obtained has been confided to specialists in Germany for study. Three papers dealing with the crystalline schists (gneisses),⁵⁸ pegmatites,⁵⁹ and granites,⁶⁰ have recently appeared. The rocks of the granite contact-zones had been placed in Professor Lessen's

hands before his death, while communications on other special groups are doubtless to be expected.

These investigations naturally suffer from the forced absence of all field observations on the part of their authors, but the purely petrographical study of the material brings to light many points of interest, while it furnishes the only sort of detailed information regarding the rocks of these remote regions which we can for the present hope for. It is here desired only to direct attention to a few of the most striking results obtained from the Brackebusch material by the three authors last cited.

Dr. Kühn's paper on the crystalline schists treats principally of gneiss, and offers little that is new. It is mostly occupied with additional evidence of structural and chemical changes due to dynamic metamorphism in the sense of Lehmann. The most noteworthy of these are development and microstructure of fibrolite; production of augen-gneiss from porphyritic granite; development of microcline structure in orthoclase by pressure; secondary origin of microcline, microperthite and micropegmatite; alteration of garnet to biotite and hornblende.

Dr. Sabersky's paper on the coarse-grained granites or pegmatites is entirely mineralogical, and is devoted principally to elucidating the structure of microcline. The author concludes that the well-known gridiron structure is due, not to two twinning laws (the Albite and Pericline), as has been generally supposed, but to the Albite law alone, in accordance with which the individuals form both contact and penetration twins, like the albite crystals from Roc-tourné, described by G. Rose.

Dr. Romberg's paper on the Argentine granites is much more extensive than the two preceding. It is embellished by seventy-two microphotographs, many of which admirably illustrate the special points described. He comes to several results of great petrographical significance, the most important of which relate to the origin of quartz-feldspar intergrowths in granitic rocks. He clearly shows that beside the original granite quartz there is also much of a secondary nature present. This is not microscopically distinguishable from the original mineral, but its later genesis is demonstrated by many careful observations on its relation to other constituents. The abundant secondary quartz is regarded as the product of weathering—principally of the feldspar, into which it has a peculiar tendency to penetrate. The extreme

sensitiveness of quartz to pressure is emphasized (as it has been by Lehmann and the present writer) and illustrated by undulatory extinction, banding, granulation and even plastic bending around other minerals. Dynamic action is regarded as the efficient cause of the secondary impregnation of feldspar by quartz, and a union of this with weathering of the feldspar as the source of the abundant and complex pegmatitic intergrowths of quartz and feldspar.

These results are important, and they will now doubtless come to be generally recognized. It is, however, of interest to observe in this connection that all which is here announced as new in regard to secondary and "corrosion" quartz was described and figured in even greater detail by Prof. R. D. Irving ten years ago. This does not appear to be known to Dr. Romberg, for he does not allude to it, but anyone who will turn to pages 99 to 124 and plates XIII, XIV and XV of the monograph on the Lake Superior Copper Rocks (vol. 5, U. S. Geol. Survey, Washington, 1883) will find his conclusions stated in almost the same language and with a much wider range of fact and illustration. Dynamic action is not here adduced as a cause for the saturation of feldspar by secondary micropegmatitic quartz, since the Lake Superior rocks show no evidence of having been subjected to pressure, but that the quartz itself has been derived from the leaching of the feldspar substance and that the impregnation is mostly confined to the orthoclase is clearly stated.

Dr. Romberg also demonstrates, in a number of cases, the secondary origin of albite, especially as microperthite, and of microcline. He gives details relating to each of the mineral constituents, and then the effects of pressure and of chemical action on the most important of them. Among many interesting observations but a few can be even mentioned here; such, for instance, as the original character of muscovite in many granites; the alteration of garnet into muscovite; the dependence of the well-known pleochroic halos in biotite and cordierite upon the substance of the zircon which they almost invariably surround, and secondary rutile needles which grow out from biotite into both quartz and feldspar. In one rock occurring in a granite a violet, strongly pleochroic mineral was found, which, in neither composition nor physical properties, agreed exactly with any known species. It seems to be intermediate between andalusite and dumortierite,

but, as its individuality is not yet perfectly established, no new name is proposed for it.

G. H. WILLIAMS.

The Mineral Industry, its Statistics, Technology and Trade, in the United States and Other Countries, from the Earliest Times to the End of 1892. Vol. I. Edited by RICHARD P. ROTHWELL, editor of the *Engineering and Mining Journal*. 629 pp., 8vo.

This volume is a statistical supplement of the *Engineering and Mining Journal*, and is published by the Scientific Publishing Co., of New York, 1893. It takes the place of the former annual statistical number of the *Engineering and Mining Journal*, and it is the first volume of a series which is to be issued annually. The object of the present volume is to make known, as soon as possible after the expiration of the year 1892, the statistics and the various conditions of the mining industry in that year and in previous years. The future volumes will, each year, bring these statistics up to date, and thus the full particulars of the mining industry will be known within a few days of the expiration of every year. The volume is a compilation of articles written by different authors, and the names of these writers are guarantee that the different subjects have been treated by authorities in the departments with which they deal. The editor himself, it is but justice to him to state, has written some of the most important parts of the volume, notably the article on the statistics of gold and silver, and his well-known familiarity with the subjects he discusses renders the reader confident of their accuracy.

The present volume is not confined to the bare presentation of figures of production and consumption of various mineral products, but it treats each individual branch of the mining industry in its various departments; and in this way the volume really represents a series of treatises on the various mining products and the methods of treating them. The production of each material is given not only for the United States but also for foreign countries; the conditions of the American and foreign markets during 1892 and previous years are discussed, while the various uses of the different materials, the history of mining in different districts, the means of transportation, the metallurgical methods of treating different ores, the

methods of sampling, and the possibilities of competition in various mining industries are also described. In addition to this, tables of assessments levied and dividends paid by various mining companies are given. The volume ends with a concise statement of the statistics and condition, as well as the extent, of the mining industries of foreign countries. Thus there is presented, in a volume of no excessive size, a complete and concise epitome of the mining industries of the world; and this work was completed almost immediately after the time to which it relates.

The various subjects are treated in the following order: A résumé and tables of statistics of the mineral products of the United States; articles on Aluminum, Antimony, Asbestos, Asphaltum, Barytes, Bauxite, Borax, Bromine, Cement, Chemical Industry, Chromium, Coal and Coke, Copper, Corundum and Emery, Cryolite, Feldspar, Fluorspar, Gold and Silver, Iron and Steel, Lead, Manganese, Mica, Nickel and Cobalt, Onyx, Petroleum, Phosphate Rock, Platinum Group of Metals, Plumbago, Precious Stones, Pyrites, Quicksilver, Salt, Soda, Sulphur, Talc, Tin, Whetstones and Novaculite, Zinc; Tables of Assessments Levied by Mining Companies from 1887–1893; Tables of Dividends Paid by American Mining Companies; Baltimore Mining Stock Market, Boston Mining Stock Market, Denver Mining Stock Market, London Mining Stock Market, Lake Superior Mining Stock Market, New York Mining Stock Market, Paris Mining Stock Market, Pittsburg Mining Stock Market, Salt Lake City Mining Stock Market in 1892, San Francisco Mining Stock Market; Foreign Countries—Austria-Hungary, Belgium, Canada, China, France, Germany, Italy, Japan, Russia, South American Countries, Spain and Cuba, Sweden, United Kingdom of Great Britain and Ireland.

The importance of the subject treated in this volume can be appreciated when it is known that the products of the mines of the United States alone in the census year of 1889 amounted to \$587,230,662, and that this amount really only represents the interest on an immensely larger capital invested. The mining products of the United States are far more important in their aggregate value than those of any other country in the world, though, in many individual products, other countries supply more than the United States. This country is first, however, in the production of pig iron and steel. It is also first in the production of copper, gold, silver, petroleum, and a number of other products. Great Britain is still the leader in the production

of coal, but the United States' production is rapidly growing and already equals 81.08% of the British production, and supplies 28.75% of the world's consumption.

Every subject in this volume is fully discussed, and at the same time nothing is given which is not appropriate and even necessary. Thus a combination of completeness and conciseness is reached which is excellent. Among the most carefully and exhaustively treated subjects are copper, gold and silver, the platinum group of metals and coal and iron, though many others might be mentioned, for every subject undertaken has been thoroughly treated. In the article on copper the statistics of production and consumption, as well as the condition of the various domestic and foreign markets, are fully discussed by the editor, and, in addition, separate articles are also given on "American Methods of Ore Sampling and Assaying," by Albert R. Ledoux, and on "Bessemerizing Copper Matte," by Charles Wade Stickney. The article on the statistics of gold and silver is by Mr. R. P. Rothwell, editor of the volume, and is an excellent piece of statistical work, giving, as it does, the statistics of production of gold and silver in the world for a number of years back. To this article are appended interesting papers on the "Chronology of the Gold and Silver Industry, 1492-1892," by Walker Renton Ingalls, on "Recent Improvements in Gold Chlorination," by John E. Rothwell, and on the "Cyanide Process," by Louis Janin, Jr.

The article on the Platinum Group of Metals, by Charles Bullman, gives complete information regarding the production, consumption, nature of the deposits, metallurgy and uses of platinum and its related metals, iridium, rhodium, osmium, palladium and ruthenium. The articles on Coal and Coke and on Iron and Steel, both by Mr. Wm. B. Phillips, give full statistics of production and consumption, as well as interesting historical data, and reports of the condition of various markets. Many of the other articles in this volume deserve mention, but lack of space forbids further detail. It may be said, however, that everything necessary is presented, and nothing unnecessary or unreliable is given; in other words, the volume contains no trash.

One of the most noticeable features of the volume is the uniform and systematic manner in which the results are presented. The uniform arrangement of statistics is a matter requiring the greatest labor and statistical ability. Compiling a single table of statistics is a simple matter,

but arranging a vast mass of statistics, relating to many diverse subjects, on a uniform and intelligible basis, is entirely another matter, and requires the highest skill of the statistician. In the Mineral Industry this has been accomplished in a most successful manner; everything is clear and intelligible at the first glance, and everything is in its proper place. A great detriment to the systematic presentation of statistics has been, as pointed out by the editor, the necessity of using our present system of weights and measures, with “our long and short tons, our barrels of 200, 280, 300 or 400 lbs, our pounds avoirdupois and our pounds Troy, our bushels of a dozen different weights, and our gallons of several incomprehensible kinds”; but the disadvantages of this system have been partly avoided in many cases by giving the statistics in metric measures as well as in our own.

The question of the cost of production has been given especial prominence in this volume, with a view to showing the reduction in the cost of the crude products. To use the words of the editor: “The itemization of cost is the first essential step in securing economy in producing any article, and the history of every country and of every industry has shown that prosperity, whether national, industrial, or individual, is, in a general way, inversely proportional to the cost of supplying the rest of the world with what one produces.” These reductions are in no way dependent on the reduction of wages. On the contrary, many of the mining industries where the greatest reduction in cost of production has been accomplished, are carried on with high priced labor; and in many other cases, where the wages are not high, the condition of the wage-earners has been greatly improved. The reduction in cost of production has been entirely brought about by improvements in mining machinery, by a more thorough understanding of the nature of the deposits to be worked, and by more intelligent management and labor. The reduction in cost of production is nowhere better seen than in the materials most necessary to our welfare. For instance, coal can in some cases be carried by rail for 400 miles and delivered on board vessels for from \$2 to \$2.25 per ton, and yet the mine owners and railroads make dividends; some of the manufacturing establishments in Western Pennsylvania obtain coal at from 60 to 75 cents per ton at their works; hard gold-bearing quartz can be crushed, washed and 95 per cent. of the gold saved on the plates for \$1.25 per ton; high grade Bessemer iron ore can be mined, handled, shipped and delivered a thousand miles from the point of production for less than \$4.00 per ton. All these

figures seem almost incredible until one investigates the various devices which the ingenuity and better education of those engaged in the industry have invented for reducing the expenses of production.

The former annual statistical numbers of the Engineering and Mining Journal were excellent in all they undertook, but the present volume, the Mineral Industry, makes a great advance in giving the statistics for foreign countries in addition to those of the United States. By so doing it gives the American producers an opportunity to know the present, past and probable future conditions of competition in foreign countries.

The two most important features in any statistical work are accuracy and promptness. The necessity of accuracy is self-evident, and without promptness the statistics lose much of their serviceability to those most interested in them, for the statistics of an industry published a year or two years late are rarely of much value to those engaged in that industry. The business man wants his statistics immediately after the expiration of the time to which they relate, so that he may know the existing condition of the industry in which he is engaged; but if he does not get these statistics until many months or even several years afterwards, the condition of the industry may have changed entirely since the time to which the statistics refer. It is the promptness with which this volume is issued, combined with a high degree of accuracy, far greater than would be expected in statistics so hastily compiled, that gives it its especial value.

In conclusion, it may be said, that as a piece of statistical work, relating to an industry that is world-wide in its scope, combining accuracy with full detail and systematic arrangement, and issued so soon after the close of the time to which it relates, the Mineral Industry has never been equaled in this country or abroad. The former statistical numbers of the Engineering and Mining Journal, which referred mostly only to American mining, were considered remarkable pieces of statistical work, on account of the promptness of their publication; but in the Mineral Industry we have an epitome of the mining operations of every quarter of the globe, published almost immediately after the close of the time to which they refer, a feat which heretofore would have been declared impossible. This accomplishment is most creditable to the editor, Mr. Rothwell, to the systematic organization of the Scientific Publishing Co., and to the business manager, Mrs. Braeunlich, by whose business ability such an expensive

undertaking is made commercially practicable. The volume will be found of the greatest value to the economic geologist, the miner, the engineer and the business man.

R. A. F. PENROSE, JR.

FOOTNOTES

- [⁵⁴](#) Geological Observations in South America, 1846.
- [⁵⁵](#) Beiträge zur Geologie und Paleontologie der argentinischen Republik; I. Geologischer Theil, 1885.
- [⁵⁶](#) Studien über Cordillerengesteine. Apolda, 1875.
- [⁵⁷](#) Reisen in den Cordilleren der argentinischen Republik, Verh. der Gesellsch. für Erdkunde. Berlin, 1891.
- [⁵⁸](#) Untersuchungen an altkrystallinen Schiefergesteinen aus dem Gebiete der argentinischen Republik, von B. Kühn. Neues Jahrbuch für Min., etc., Beit. Bd. VII., 1891, p. 295.
- [⁵⁹](#) Untersuchung argentinischer Pegmatite, etc., von P. Sabersky, *ib.*, p. 359.
- [⁶⁰](#) Untersuchungen an argentinischen Graniten, etc., von J. Romberg, *ib.*, VIII., pp.

ANALYTICAL ABSTRACTS OF CURRENT LITERATURE. [61](#)

***A New Tæniopteroid Fern and its Allies.* By DAVID WHITE.
(Bulletin Geological Society of America, 4 pp., 119–122, pl.
I.).**

Mr. White has described, under the name of *Tæniopteris missouriensis*, a new and well characterized fern from the Lower Coal-measures in the vicinity of Clinton, Henry County, Missouri. Botanically, it is of particular interest in that it combines the so-called tæniopteroid and alethopteroid types of structure, while geologically it is of much value in supplying a readily identified stratigraphic mark in a part of the Carboniferous not especially rich in fossil plants. After thoroughly describing it and considering its specific and generic resemblances, the author discusses at length its suggested genetic relations and represents in a graphic manner a scheme of its probable ancestors and line of descent.

F. H. K.

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***Rainfall Types of the United States.* Annual Report by Vice-
President GENERAL A. W. GREELY. (The National
Geographic Magazine. Vol. V., April 29, 1893, pp. 45–58 pl.
20).**

The paper confines itself to the characteristic distribution of precipitation throughout the year and gives the rainfall types of the country.

(a) The best defined type of rainfall within the United States is that which dominates the Pacific coast region as far east as western Utah. The characteristic features are a very heavy precipitation during midwinter, and an almost total absence of rain during the late summer. (b) The characteristics of

the Mexican type, dominating Mexico, New Mexico and western Texas, are a very heavy precipitation after the summer solstice and a very dry period after the vernal equinox. August is the month of greatest rainfall, while February, March and April are almost free from precipitation. (c) The Missouri type covers the greatest area, dominating the watersheds of the Arkansas, Missouri and upper Mississippi rivers, and of lakes Ontario and Michigan. It is marked by a very light winter precipitation, followed in late spring and early summer by the major portion of the yearly rain, the period when it is most beneficial to the growing grain. (d) The Tennessee type, prevailing in Kentucky, Tennessee, Arkansas, Mississippi and Alabama, has the highest rainfall the last of winter, while the minimum is in mid-autumn. (e) The Atlantic type, covering all the coast save New England, is one where the distribution throughout the year is nearly uniform, with a maximum precipitation after the summer solstice, and a minimum during mid-autumn. (f) The St. Lawrence type is characterized by scarcity during the spring months, heavy rainfall during the late summer and late autumn months, with a maximum during November.

The regions lying between these several type-regions have composite rainfall types, resulting from the influence of two or more simple types.

H. B. K.

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***The Geographic Development of the Eastern Part of the
Mississippi Drainage System. By LEWIS G. WESTGATE,
Middletown, Conn. (American Geologist, Vol. XI, April,
1893, 15 pp.)***

The drainage of the Eastern Mississippi basin in post-carboniferous was in all probability consequent upon the tilting which accompanied the stronger folds of the Appalachian revolution in the east. The present drainage is found to accord in the main with this hypothetical post-carboniferous drainage, but several streams depart quite widely from it.

(a) The great drainage lines of the St. Lawrence basin are structural valleys developed along the strike of the softer Paleozoic strata, and at right angles to the original surface. The streams seem, therefore, to have adjusted themselves to the differences in hardness and structure of the beds discovered.

(b) The Ohio and Cumberland rivers cut directly across the Tennessee and Cincinnati anticlines. The most probable explanation is that the rivers were superimposed upon the arched and eroded Silurian rocks from a thin cover of carboniferous beds—now entirely removed. (c) The Upper Mississippi does not follow the dip of the rocks to the southwest, but follows the strike to the southeast. This part of the river probably dates from the elevation of the plains on the west and the Appalachians on the east, which marked the close of the Cretaceous and which left a broad north and south valley. (d) The author finds good reason to believe that the Lower Mississippi, in post-carboniferous times, flowed west through Missouri and Arkansas. The present course was probably taken at the close of the Cretaceous in consequence of elevations on the west and east, and possible depression in the south.

The Cretaceous base-level recognized by Davis on the Atlantic slope can be traced more or less discontinuously, and remnants of it are believed to exist in Kentucky, Tennessee, Wisconsin, Minnesota and Arkansas. But in general the work of the Tertiary cycle has obliterated almost all evidence of it on all but the hard sandstones and conglomerates of the Paleozoic series.

Good examples of the lowlands excavated from the Cretaceous base-level during the Tertiary cycle, are the Valley of the East Tennessee and the central lowland of Kentucky and Tennessee. During the post-Tertiary sub-cycle the larger streams trenched to greater or less extent these lowlands. No attempt is made to carry the history of the development of the Mississippi, drainage into the complicated chapter of the ice-invasion.

H. B. K.

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***On a New Order of Gigantic Fossils.* By ERWIN H. BARBOUR.
(University Studies. Published by the University of
Nebraska. Vol. I, No. 4, July, 1892, pp. 23, pl. 5).**

A part of Sioux County, Nebraska, lying north of the Niobrara River, has yielded a new order of gigantic Miocene fossils unlike anything heretofore known. They are best described as fossil corkscrews, of great size, coiling in right-handed or left-handed curves about an actual axis or around an imaginary axis. The screws are often attached at the bottom to an immense transverse

piece, rhizome, underground stem, or whatever it may be, which is sometimes three feet in diameter. In other cases the corkscrew ends abruptly downward, as it always does upwards. In still other cases the transverse piece is variously modified, and sometimes blends into the sandstone matrix, as if the underground stem, while growing at one end, was decaying at the other. The fossil corkscrew is invariably vertical, and the so-called rhizome invariably curves rapidly upwards, and extends outwards an indefinite distance.

That they could ever have been formed by burrowing animals, by geysers or springs, or by any mechanical means whatever, is entirely untenable. Their organic origin is unquestionable. Microscopic sections show smooth spindle-shaped rods, which are suggestive of sponge spicules. From the numbers seen in place it is evident that they flourished in thickly crowded forests of vast extent.

A finely preserved rodent's skeleton was found in one great stem. The probable explanation is not that the rodent burrowed there, but that its submerged skeleton became an anchorage for a living, growing *Daimonelix*, which eventually enveloped it.

The author proposes this provisional classification:

Order.	Family.	Genus.	Species.
	{ <i>Daimonelicidæ</i> .	{ <i>Daimonelix</i> .	{ <i>circumaxilis</i>
	{	{	{ <i>bispiralis</i>
_____	{ _____	{ _____	{ <i>anaxilis</i>
	{	{	{ <i>robusta</i>
	{	{	{ <i>carinata</i> .
		{ _____	{ _____
		{ _____	{ _____

The different species are described in full.

H. B. K.

***The Vertical Relief of the Globe.* By HUGH ROBERT MILL,
D.Sc., F.R.S.E. Scottish Geographical Magazine, April,
1890.**

The purpose of Dr. Mill's paper is to show a simple yet adequate basis on which to build the superstructure of physical geography. It does not attempt a discussion of the distribution and varieties of vertical relief. The structure of the earth is stated most simply by describing it as an irregular stony ball, covered with an ocean and an envelope of air. If the lithosphere were perfectly smooth and at rest, with the hydrosphere uniformly spread over its surface, the former would have the form of the terrestrial spheroid, and the latter would surround it to a depth of 1.7 miles. The surface of this hypothetical spheroid Dr. Mill calls *mean sphere level*. Of course, in reality the surface of the lithosphere is not perfectly smooth. Parts of it are greatly depressed and parts much elevated, the latter forming the land of the earth. The writer proceeds to calculate the position of mean sphere level, and in the absence of accurate data he uses the careful estimates of Dr. John Murray, which are as follows: Average depths of oceans = 2.36 miles; average height of land = .426 miles; average thickness of hydrosphere surrounding smoothed lithosphere = 1.7 mile; area of land = 55,000,000 square miles; area of oceans = 141,700,000 square miles. Suppose a block of 55,000,000 of square miles area and 1.7 miles deep to be cut out of the smoothed lithosphere and set down on the surface alongside the depression. No change will take place in the surface of the hydrosphere. If the surface of the 141,700,000 square miles of lithosphere were reduced to uniformity, the whole depressed area would lie .66 mile beneath mean sphere level, and the depth of the ocean becomes 2.36 miles. To raise the land to its actual mean level above the hydrosphere surface, a sufficient quantity of matter must be removed from the depressed area and placed on the elevated block. Let x = the thickness of the belt removed and y equal the thickness of the belt when placed on the elevated block. Then $x + y$ is the height of the land above the actual hydrosphere level. From the data given the following equations are easily obtained:

$$\begin{array}{rclcl} x + y - .426 & = & 0 \\ 141.7 x - 55 y & = & 0 \\ x & = & .12 \text{ and } y = .306 \text{ in miles.} \end{array}$$

The average height of the land above mean sphere level is thus $1.7 + .306 = 2.006$ miles, and the average depth of the depressed portion beneath mean sphere level is $.66 + .12 = .78$ mile.

Dr. Mill divides the earth into the three following divisions: (1) *Abysmal area*, occupying all the depressions beneath the mean surface of the lithosphere, occupying 50 per cent. of the earth's surface; (2) *Transitional area*, comprising all the regions above mean sphere level covered by the hydrosphere, occupying 22 per cent. of the surface; (3) *Continental area*, all the lithosphere that projects above the hydrosphere, or 28 per cent. of the earth's surface.

J. A. B.

FOOTNOTE

⁶¹ Abstracts in this number are prepared by F. H. Knowlton, Henry B. Kummel, J. A. Bownocker.

ACKNOWLEDGMENTS.

The following papers have been donated to the library of the Geological Department of the University of Chicago:

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(Further acknowledgments of pamphlets already received will be made in the next number.)

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Footnotes have been sequentially renumbered to make them unique, and then moved to the ends of the articles to which they belong.

Illustrations and references to them have been sequentially renumbered.

Pages [344](#), [345](#): Text shown as boldface was printed in a Blackletter font.

[Footnote 30](#), originally on page [368](#): The page range was printed as shown (with dashed placeholders); the missing numbers may be 369 and 370.

Transcriber added a footnote link to the caption of [Figure 2](#), as the footnote explains the abbreviations used in the illustration.

[Footnote 46](#), originally on page [382](#): The citation for "Proc. Bos. Soc. Nat. Hist." had no further details.

Page [411](#): Text uses both "Schiefergesteinen" and "Schiefergestenien".

[Footnote 60](#), originally Footnote 7 on page [411](#), omitted the page number reference.

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